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REVIEW

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Acidic oxygen evolution reaction: Mechanism, catalyst classification, and enhancement strategies

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Abstract

As the most desirable hydrogen production device, the highly efficient acidic proton exchange membrane water electrolyzers (PEMWE) are severely limited by the sluggish kinetics of oxygen evolution reaction (OER) at the anode. Rutile IrO_2 is a commercial acid-stable OER catalyst with poor activity and high cost, which has motivated the development of alternatives. However, hitherto most of the designed acidic OER catalysts have disadvantages of low activity or stability, which cannot meet the requirement of industrial applications. Thus, exploring suitable strategies to enhance the activity and stability of cost-effective acidic OER catalysts is crucial for developing the PEMWE technique. In this review, the main OER mechanisms, different types of catalysts, and their activity and stability characteristics are summarized and discussed, and then possible strategies to improve activity and stability are proposed. Finally, the problems and prospects of such catalysts are generalized to shed some light on the future research of advanced catalysts for acidic OER.

KEYWORDS

catalyst, hydrogen energy, oxygen evolution reaction, proton exchange membrane water electrolyzer, water oxidation

1 | INTRODUCTION

The scarcity of resources and environmental damage make it increasingly important to develop environmentally friendly and sustainable energy sources.[1,2] Hydrogen is an ideal candidate for the next energy due to its high energy density and cleanliness.[3,4] However, the production of gray and blue hydrogen still inevitably causes carbon emissions, which is not conducive to achieving carbon peaking and carbon neutrality goals.[5] By comparison, green hydrogen produced by electrocatalytic water splitting has the advantage of decarbonization, which is in line with the global development target of carbon dioxide emissions reduction.[6] Moreover, the hydrogen from water electrolysis can be used as an energy carrier to store electricity generated by wind turbines and photovoltaics.[7]

Currently, two main types of water electrolysis devices are alkaline anion exchange membrane water electrolyzers (AEMWE) and acidic proton exchange membrane water electrolyzers (PEMWE).[8,9] The ion mobility and diffusion coefficient of OH^- in anion exchange membrane is much lower than that of H^+ in proton exchange membrane (PEM), which allows PEMWE with higher conductivity and lower ohmic loss.[10] Moreover, hydrogen gas is difficult to pass through PEM and enter the anode side, which can control the mixture of hydrogen and oxygen in a safe range, thus ensuring the safe operation of the water electrolysis device.[11,12] Therefore, the cathode of PEMWE can operate in an environment of high-density hydrogen generated by working at high voltage, while avoiding the risk of explosion of the device due to excessive hydrogen transfer into the anode.[11,12] Moreover, the stacked configuration of PEMWE can be simplified by the ability to operate at high pressures. The proton transport through PEM has a fast response to voltage, so PEMWE have a faster system response than AEMWE.[13] Therefore, PEMWE are more promising for hydrogen production because of its higher voltage efficiency and current density, with greater safety and simpler installation than AEMWE.

Water oxidation, or oxygen evolution reaction (OER), is a complex process involving a multistep proton-coupled four-electron transfer.[14–16] Its slow kinetics process requires a high voltage to drive, which limits the efficiency of water splitting.[17,18] In particular, the process of OER is even slower in acidic environments.[19,20] Consequently, the inherent high potential of the OER process in PEMWE and the severe corrosion under a strong acidic environment make the long-term operation of the catalyst face huge challenges.[21] The appropriate degree of antiorbital filling and the level of orbital hybridization allows Ir- and Ru-based materials with relatively high acidic OER activity.[22–24] However, hitherto, the only catalyst that can efficiently drive acidic OER for long periods of time is rutile IrO_2, but its low earth abundance and poor activity make it imperative for the hydrogen industry to move away from its over-reliance on Ir.[25] RuO_2, another commercial acidic OER catalyst, has a relatively cheap price and high activity but is easily deactivated by oxidation to soluble RuO_4.[26] Thus, as promising alternatives to Ir (Ru) based materials, nonprecious metal- and carbon-based acidic OER catalysts have been reported, but their difficult combination of activity and stability makes them unsuitable for commercial applications.[27] Therefore, the rational design of acidic OER catalysts with higher activity and stability is the sticking point to further advance the larger-scale hydrogen production from PEMWE.[12]

This review presents recent research progress on acidic OER (Figure 1), which is expected to guide the next step in the development of such catalysts. First, the understanding of the two classical reaction mechanisms is discussed and the characteristics of the different mechanisms are highlighted. Next, we provide a detailed overview of the three acidic OER catalyst types to date, including precious metal-, nonprecious metal- and carbon-based catalysts. Then, strategies for promoting the activity are summarized, mainly focusing on seven aspects at least: composition, substrate, morphology, atom-doping, defect, strain, and anion engineering. Furthermore, five strategies at least for excitation stabilization are introduced, which are wrapping, enhancing electronic coupling, stabling lattice oxygen, electronic structure regulation, and self-healing. Finally, according to the current understanding of acidic OER catalysts, challenges, and feasibility prospects are proposed.

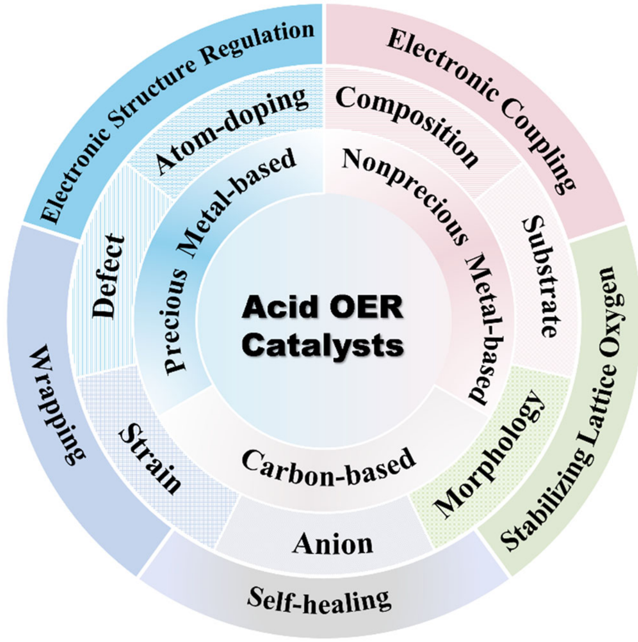

FIGURE 1 Reaction mechanisms, main types, strategies to promote activity and stability of acidic oxygen evolution reaction catalysts.

2 | REACTION MECHANISM

An in-depth understanding of the reaction mechanism is extremely critical for the construction of OER catalysts with good activity and stability. So far, the dominant OER mechanisms in acidic environments mainly include the adsorbate evolution mechanism (AEM) and lattice oxygen mechanism (LOM).

2.1 | AEM

The AEM in acidic media is the process that two water molecules adsorbed at the active site are converted to oxygen and released after four consecutive proton–electron transfer processes via OH*, O*, and OOH*.[28] The specific process is as follows (Figure 2A). First, the water molecule

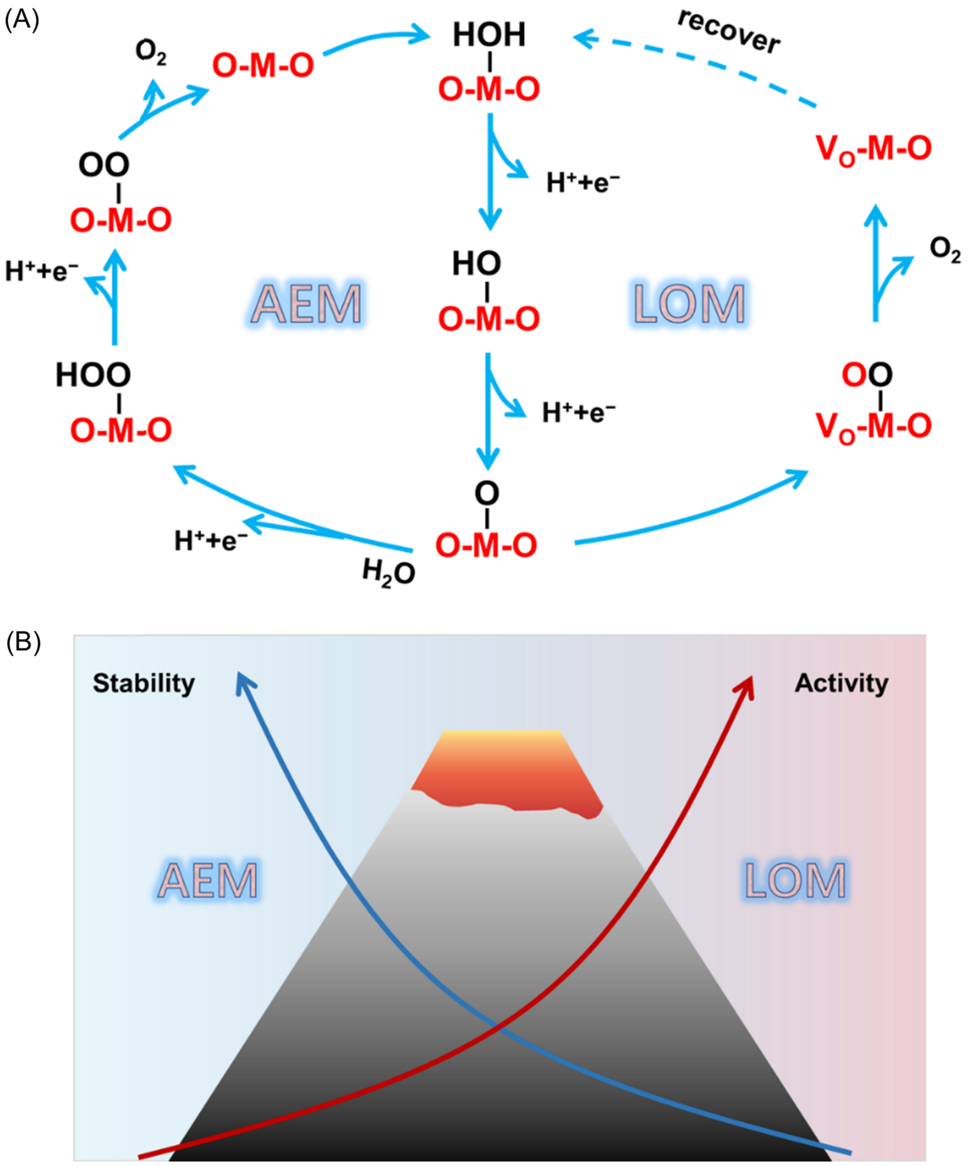

FIGURE 2 (A) Schematic diagram of the reaction mechanism and (B) activity and stability of different mechanisms. AEM, adsorbate evolution mechanism; LOM, lattice oxygen mechanism.

is adsorbed at the active site. After two successive deprotonation processes, it is then transformed into O*. Another water molecule produces a nucleophilic attack, leading to the construction of an O–O bond and the formation of a HOO* intermediate. The oxygen molecule is formed from HOO* by the last deprotonation process and released from the active site.

$$H_2O^* \rightarrow OH^* + H^+ + e^-,$$

$$OH^* \rightarrow O^* + H^+ + e^-,$$

$$O^* + H_2O \rightarrow OOH^* + H^+ + e^-,$$

$$OOH^* \rightarrow O_2 + H^+ + e^-.$$

In the same way, as in alkaline media, the AEM in acidic environments includes the same adsorbed oxygen-containing intermediates (OH*, O*, and OOH*).[29,30] The binding energy of the active site to the oxygen-containing intermediates determines the free energy of every step, in which having the maximum free energy is the rate-determining step (RDS).[31] Therefore, the binding strength of the active site to the oxygen-containing intermediates is the sticking point to determine the activity of the catalyst. The activity can be evaluated by the free energy difference (ΔG) of different intermediates. In the case of acidic OER catalysts, the RDS is often deprotonation of OH* or O–O coupling.[24,32] According to Sabatier's principle, for an ideal catalyst the oxygen binding energy should be neither too strong nor too weak. The ΔG_{O*}–ΔG_{OH*} can be used for the definition of the oxygen binding capacity to give a reasonable estimation of the catalytic activity.[33]

However, AEM provides a limited interpretation of the source of catalytic activity. Theoretically, there is the scaling relationship between OH* and OOH*, that is, $\Delta G_{OOH*} = \Delta G_{OH*} + 3.2 \pm 0.2$ eV. This makes the water oxidation process require a higher theoretical overpotential (370 mV), which cannot explain the mechanism of many highly active acidic OER catalysts.[34,35] Moreover, the structure of the catalyst is relatively stable according to the characteristics of AEM. However, many studies point out that the composition and structure of catalysts undergo significant changes during the reaction.[36,37] Therefore, AEM alone is not sufficient to make predictions about the properties of OER catalysts.

2.2 | LOM

The source of oxygen molecules generated at the active site can be explored by isotopic labeling and differential electrochemical mass spectrometry, confirming that part of the lattice oxygen is involved in the OER process.[38] The mechanism by which this lattice oxygen is oxidized and thus involved in the OER process is known as LOM (Figure 2A). First, similar to the first two steps of AEM, the adsorbed H_2O undergoes a transformation into O* after two successive deprotonations. O* and a lattice oxygen undergo coupling to form an oxygen molecule and an oxygen vacancy:

$$H_2O + * \rightarrow OH* + H^+ + e^-,$$

$$OH* \rightarrow O* + H^+ + e^-,$$

$$O* + O_L \rightarrow O_2 + Vo.$$

It has also been shown that at the beginning of the reaction, O_2 can be formed directly between the lattice oxygen atoms in the oxide by direct coupling[39]:

$$2O_L \rightarrow O_2 + 2Vo.$$

The formed oxygen vacancies can then be recovered by a refilling process.[40] Oxygen atoms in water molecules fill the oxygen vacancies and undergo a deprotonation process to form adsorbed hydrogen. Finally, the adsorbed hydrogen is removed to form a clean active site.

$$Vo + H_2O \rightarrow O_L + H* + H^+ + e^-,$$

$$H* \rightarrow * + H^+ + e^-.$$

Unlike the coordinated proton–electron transfer process in AEM, LOM exhibits a proton-electron transfer that does not coincide.[41] This leads to a significant pH-dependence of LOM.[41] Moreover, the choice of the OER mechanism is related to the applied voltage. For example, when voltages are less than 1.12 V versus SCE, the OER process for $Ru_{0.9}Ni_{0.1}O_{2-\sigma}$ takes the form of AEM.[42] While at voltages above 1.12 V versus SCE, the lattice oxygen is oxidized.[42] Furthermore, the scope of LOM tends to involve only the surface and the near-surface region of metal oxides.[43]

In fact, in the actual OER process, there is often a coexistence of AEM and LOM with some degree of competition (Figure 2B). The LOM shows a lower theoretical voltage due to the direct binding of lattice oxygen to oxygen-containing intermediates, which breaks the limits of AEM.[44,45] Therefore, the LOM usually offers a more prominent catalytic activity than AEM. Unlike the more stable active center in AEM, the structure of the catalyst in LOM is always in dynamic change during the reaction.[23,46] Even though LOM can contribute significantly to the catalytic activity, it will seriously risk the electrochemical stability of the catalyst.[23,46] Specifically, LOM leads to the formation of oxygen vacancies and accelerates the dissolution of metals, which can trigger the collapse of the catalyst structure and lead to rapid catalyst deactivation.[47] The AEM, on the other hand, does not involve the depletion of lattice oxygen and therefore has a more stable lattice.[48]

3 | MAIN TYPES OF ACIDIC OER CATALYSTS

The great significance of acidic OER catalysts for the hydrogen production industry has prompted a wide range of materials to be explored. Here, Figure 3A shows the two key indicators of catalyst performance, including the overpotential corresponding to 10 mA cm^{-2} (η_{10}) and stability test time for a variety of catalysts reported in recent years. This graph shows only general trends in the performance of various acidic OER catalysts, and not all tests are based on the same environment. Furthermore, according to the main components of the material, acidic OER catalysts can be divided into three types: precious metal-, nonprecious metal-, and carbon-based materials. As for the

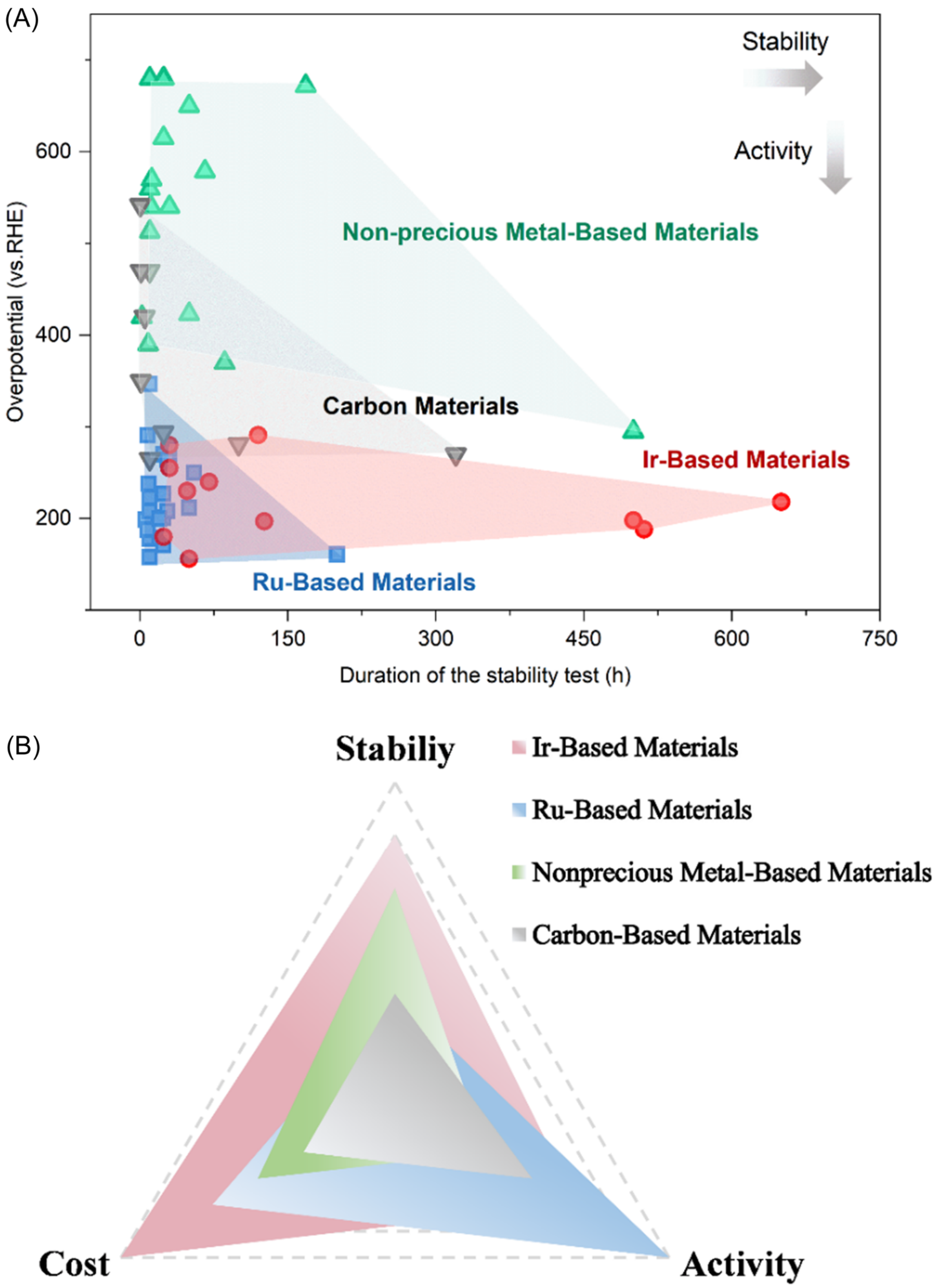

FIGURE 3 (A) Stability test time and η_{10} of catalysts reported in recent years and (B) the stability, activity, and cost of Ir-, Ru-, nonprecious metal-, and carbon-based materials.

precious metal catalysts, we labeled these two types separately in Figure 3A because Ir- and Ru-based materials have different characteristics, and each developed into larger species. As shown in Figure 3A, Ru- and Ir-based materials are distributed at the bottom of the figure, demonstrating their excellent catalytic activity. Nonprecious metal and carbon-based materials are relatively inactive and therefore have poor energy conversion efficiency. Additionally, the Ir-based material has the widest span along the *X*-axis, showing its best potential in terms of stability. Although it seems that Ir-based materials are the best option for acid OER catalysis, the extremely high cost is its fatal problem. Therefore, the characteristics of different catalysts in terms of cost and performance are further summarized in Figure 3B. In general, various catalysts have their own advantages and disadvantages in terms of stability, activity, and cost. Here, the specific characteristics of precious metal-, nonprecious metal-, and carbon-based catalysts are elaborated.

3.1 | Precious metal-based catalysts

3.1.1 | Ir-based catalysts

Ir-based materials are considered to be the benchmark for acidic OER because they can be operated for long periods in strongly acidic media. However, the annual production of Ir is extremely low and its cost is extraordinarily high, even higher than Pt. The two main types of Ir-based catalysts to acidic OER include iridium oxide and iridium (0-valent). Since Ir is also converted to oxide species under acidic OER conditions, the nature of iridium oxide is discussed first here.

The commercial rutile IrO_2 is less active in acidic media, resulting in lower energy conversion efficiency. Amorphous iridium oxide (IrO_x) has been shown to possess higher catalytic activity than crystalline IrO_2, which may be attributed to more active sites involved in the reaction and more flexible structures.[49,50] Furthermore, there are two binding energies of Ir species in amorphous iridium oxide.[51,52] The lower binding energy species corresponds to Ir(IV) in rutile IrO_2.[51] On the other hand, the O 2p hole state induced by the Ir(III) species with higher binding energy leads to the generation of chemoelectrophilic oxygen O^{I-} species, which would aggravate the nucleophilic attack on the OER process.[51] Mayrhofer's group confirmed that the presence of Ir(III) contained in hydrated iridium oxide can promote the precipitation of lattice oxygen and thus accelerate the dissolution of Ir.[53] Jones et al.[54] suggested that the crystallinity of iridium oxide would directly determine the extent of deprotonation, thereby affecting the activity and stability of the catalyst. IrO_2 with good crystallinity undergoes deprotonation only at the surface, and IrO_x with amorphous and lower crystallinity can undergo deprotonation in the near-surface region.[54] This subsurface deprotonation reduces the barriers to OER through electron doping effects but also exacerbates the dissolution of Ir.[54]

The surface layer of Ir is inevitably oxidized to IrO_x species to continue driving the reaction under acidic OER conditions.[55] The electrochemically formed iridium oxide has a higher solubility than the thermally oxidized product due to its poor thermodynamic stability.[56] There are two possible routes for the dissolution of Ir (0), including direct dissolution to Ir (III) and first oxidation to Ir(V) followed by reduction to Ir(III).[57] Maillard et al.[58] proposed that the formation of iridium oxide species is independent of the total surface free energy. Both the initial and lower total surface free energy of Ir (210) single crystal surfaces eventually formed Ir(IV) in the oxide species, and thus reduced the electrochemically active surface area and catalytic activity. This is because in the acidic environment, the surface state of the catalyst can change greatly after the OER process so the OER activity does not show a significant correlation with the state of the catalyst surface. Some studies have shown that the products obtained by electrochemical oxidation of Ir with different crystal plane orientations are different. For example, Li et al.[59] demonstrated that high-index open planes such as Ir (024) formed metastable metastoichiometric IrO_x species in the subsurface with higher catalytic activity. However, the topmost closed planes such as the (113) plane would be transformed into oxides such as IrO_2 with good stability and might reduce the catalytic activity.

3.1.2 | Ru-based catalysts

Although Ir-based materials can work stably over long periods of time when driving acidic OER, they are extremely expensive and scarce. Among the noble metal-based acidic OER catalysts, Ru has the highest catalytic activity except Os, which is exceedingly unstable and has low Faraday efficiency.[60] Moreover, the price of Ru is only one-tenth of that of Ir.[61] Therefore, Ru-based catalysts exhibit cheaper prices and stronger catalytic activity compared to Ir-based materials. However, Ru-based materials are always constrained by serious stability problems.[62] In an acidic OER environment, Ru-based materials tend to dissolve after forming high-valent ruthenium oxides (RuO_4), which leads to serious performance degradation problems.[26,62]

Ru monomers exhibit a stronger acidic OER activity than their oxides.[56] At a potential lower than that required for OER (0.8 V vs. RHE), a layer of hydrated oxide can form on its surface.[56] The high defect density of this electrochemically synthesized oxide can lead to an intensification of LOM, thus making the dissolution rate of Ru several orders of magnitude higher than that of the oxide obtained by its thermal oxidation.[60] Furthermore, the corrosion and OER processes of Ru have the same intermediates, and both processes tend to occur at the same potential.[62] The activity and stability of Ru are related to its crystal plane index. For example, Xia et al.[63] proved that Ru (111) has a superior OER activity than Ru (110). However, Ru monomers with a low index surface present stronger stability.[64,65] Polycrystalline Ru shows increased metal solubility due to its richness in high-energy defect atoms.[66] In contrast, only Ru atoms located at the edges of Ru monomer with low-index surfaces are easily soluble in the electrolyte.

RuO_2, one of the most classical OER catalysts, has high catalytic activity and poor stability under acidic

OER conditions. The activity of RuO_2 shows a high correlation with the density of Ru atoms with unsaturated coordination (Ru_{CUS}). Yang et al.[67] proposed that Ru_{CUS} has a more optimized reaction pathway than Ru with saturated coordination in catalytic OER. First, the water molecules adsorbed at the active site undergo two consecutive deprotonation processes to become fully oxidized surfaces. Then, the second water dissociation process leads to the formation of the −OO structure stabilized by the action of the adjacent −OH. Subsequently, the oxygen molecule is released after the last deprotonation step. This optimized reaction pathway confers an even better OER activity to the catalyst. However, theoretical calculation shows that these ligand unsaturated sites are more soluble than their bridge sites, which would seriously threaten the durability of the catalyst.[68] On the other hand, it has also been proposed that the activity in some ruthenium oxides can be provided by oxygen vacancies.[47] Oxygen vacancies acting as active sites can avoid the solubility of metal sites due to excessive oxidation.[47] Moreover, similar to the amorphous IrO_x species, amorphous RuO_2 can also exhibit better acidic OER activity due to its more flexible structure.[69] Nevertheless, the reports on amorphous RuO_2 are still relatively scarce.[69]

3.1.3 | Other precious metal-based catalysts

Besides Ir and Ru, there are still some reports on other precious metal-based acidic OER catalysts. It is generally believed that the active arrangement of precious metals is $Os \geq Ru > Ir > Rh > Pt > Au$, while the stability arrangement is $Pt > Rh > Ir > Au \geq Ru \geq Os$.[60,70,71] Although Os is more active, it is more prone to severe oxidation and dissolution, resulting in a lower Faradaic efficiency (5%).[60] Therefore, Os can hardly be called a valuable OER catalyst. Pt-based materials are inevitably oxidized to PtO or PtO_2 at OER potential, and thick Pt oxides with poor conductivity can seriously threaten the OER activity of catalysts.[72] In addition, Au has slow kinetics and a high dissolution rate under acidic OER conditions, which is far from sufficient for commercial use.[39,73] Therefore, currently, Rh- and Pd-based materials are the major components of acidic OER catalysts apart from Ir- and Ru-based materials.

Rh has good electrochemical activity, corrosion resistance, and stability, showing good potential for acidic OER applications.[74,75] For example, the hollow RhCu nanowires were prepared by a sacrificial template method and were confirmed to have good acidic OER activity.[74] This Rh_3Cu_1 alloy material corresponded to an overpotential of 345 mV at 10 mA cm^{-2} and could maintain at this current density for 12 h without significant decay. Li et al.[75] reported that Rh_2P could achieve driving OER in acidic environments. The activity and stability of Rh_2P did not have significant advantages. Rh_2P required a voltage of 0.51 V to drive a current density of 5 mA cm^{-2} and can only operate for 1000 s. Apart from Rh, Pd has a low solubility rate in a high-potential acidic environment and thus is one of the promising catalysts for acidic OER.[76,77] A nanoflower-like Pd/$NiFeO_x$ catalyst that can be used for acidic OER was reported.[76] The Pd/$NiFeO_x$ nanosheets in 0.5 M H_2SO_4 required an overpotential of 169 mV to achieve 10 mA cm^{-2} and could realize long-term operation for 50 h. Recently, Wu et al.[77] constructed a Pd nanoparticle attached to a Pd backbone by electrochemical etching, demonstrating good acidic OER activity. This graded Pd material required an overpotential of 196 mV to drive a current density of 10 mA cm^{-2} and could work continuously for up to 500 h. The lattice strain on the Pd (111) surface tuned the adsorption of the active sites to the reaction intermediates, thus enhancing the catalyst activity. Except for the above, it has been reported that the deactivation of Pt particles due to oxidation to Pt oxides with low conductivity can be changed by constructing Pt single-atom materials.[78] We will elaborate on this part in the section about carbon materials.

3.2 | Nonprecious metal-based catalysts

3.2.1 | Monometallic catalysts

Considering the natural scarcity of noble metals, the exploration of nonprecious metal-based materials with great earth-abundance, activity, and stability is also particularly critical. As the most promising nonprecious metal-based OER catalysts in acidic environments, monometallic oxides, including manganese oxides, iron oxides, and cobalt oxides, have been explored.[27,79–82] For example, Schaak et al.[80] prepared a cobalt oxide film with good acidic OER activity using fluorine-doped tin oxide (FTO) as a substrate. The catalyst required an overpotential of 570 mV to achieve a current density of 10 mA cm^{-2} and could operate for 12 h in 0.5 M H_2SO_4. Besides, an iron oxide thin film composed of magnetite and hematite was reported to have certain OER activity in an acidic environment.[83] This iron oxide material in 0.5 M H_2SO_4 could achieve 10 mA cm^{-2} at an overpotential of 650 mV and required an overpotential of 886 mV after 24 h operation at this current density. Compared to cobalt oxide and iron oxide, the Mn-based oxide has a stable potential window under acidic OER conditions, thus exhibiting the possibility to drive acidic OER continuously for a long time. It has been reported that γ-MnO_2 can operate continuously in 10 mA cm^{-2} for up to 8000 h at pH 2.[82] Despite high stability, γ-MnO_2 had unsatisfactory activity (480 mV@10 mA cm^{-2}).

Aside from monometallic oxides, many monometallic compounds have been reported for application in acidic OER catalysts. For example, a serious of Si-based compounds with Fe, Co, and Mn components, respectively, could be used to drive acidic OER.[84] The prepared Fe_5Si_3, Mn_5Si_3, and CoSi required overpotentials of 0.73, 0.78, and 0.80 V, respectively, to reach the current density of 10 mA cm^{-2}. Moreover, partial transition metal chalcogenides not only have certain hydrogen evolution reaction (HER) activity but also have the potential to act as OER catalysts in acidic environments. For example, Lou et al.[85] compared the OER performance of MoS_2 and TaS_2 sheets with 1T and 2H crystalline forms, respectively, and showed that the acidic OER performance was ranked: 1T-MoS_2 > 1T-TaS_2 > 2H-MoS_2 > 2H-TaS_2. The most active 1T-MoS_2 required an overpotential of 460 mV to realize the current density of 10 mA cm^{-2} and sustained 120 min of continuous operation. It has also been reported that borides can be used as acidic OER catalysts. A TiB_2 film constructed by Dasog et al.[86] could achieve a current of 10 mA cm^{-2} with an overpotential of 560 mV and operate for more than 10 h with minimal decay.

3.2.2 | Polymetallic catalysts

The compositional diversity of polymetallic catalysts provides more flexibility in the tuning of catalyst performance.[87] Similar to monometallic catalysts, polymetallic catalysts are mainly focused on the type of oxides. For example, Lewis et al.[88] constructed a crystalline solid solution of $NiSb_2O_6$ and $MnSb_2O_6$ ($Ni_xMn_{1-x}Sb_{1.6-1.8}O_y$). The stable rutile crystals allowed the catalyst to operate continuously in 1.0 M H_2SO_4 at a current density of 10 mA cm^{-2} for 168 h, exhibiting excellent durability. Compared with $MnSb_{1.7}O_y$ (806 ± 9 mV) and $NiSb_{1.8}O_y$ (727 ± 20 mV), $Ni_xMn_{1-x}Sb_{1.6-1.8}O_y$ had the best activity with an overpotential of 672 ± 9 mV corresponding to 10 mA cm^{-2}. This also confirmed that the combination of $NiSb_2O_6$ and $MnSb_2O_6$ improved the catalytic activity. Additionally, a self-healing cobalt–iron–lead oxide ($CoFePbO_x$) has been reported to drive OER in acidic environments for long periods of time.[89] The overpotential of $CoFePbO_x$ corresponding to 1 mA cm^{-2} was 220 mV higher than that of IrO_x in an environment of pH 2.5. $CoFePbO_x$ could output 1 mA cm^{-2} at about 1.8 V for more than 50 h in an environment of pH 2.

Some nonprecious metal alloys have also been reported to have relatively good catalytic activity for acidic OER. For example, a series of intermetallic alloys (Ni_2Ta, Co_2Ta, and Fe_2Ta) prepared by arc melting were constructed by Schaak et al.[90] Compared to the pure Ni samples, Ni_2Ta had a significantly lower Ni dissolution rate, which was due to be protected by the formed tantalum oxide. However, the performance of the best-performing Ni_2Ta was also unsatisfactory, it required an overpotential of 980 mV to reach 10 mA cm^{-2}. Besides, polymetallic transition metal sulfides have been shown to have good acidic OER activity. For example, a Co-doped MoS_2 enabled the driving of OER in acidic environments.[91] The doped Co tuned the electronic structure, improved the intrinsic conductivity, and acted as an active site for OER. Thus, this Co-MoS_2 for acidic OER required an overpotential of 540 mV for an output of 10 mA cm^{-2}. A graded $MoS_2/Co_9S_8/Ni_3S_2/Ni$ for efficient OER was constructed by Kanatzidis et al.[92] The synergistic interaction and electron transfer between the multiple phases led to excellent OER activity in all-pH environments. In particular, the material required an overpotential of 228 mV to achieve a current density of 10 mA cm^{-2} in acidic media. In addition to sulfides, polymetallic transition metal phosphides have also been reported for acidic OER. For example, amorphous NiFeP bulk material for OER was prepared by Xiong et al.[93] by melt spinning and rapid quenching. The catalyst had good electrical conductivity and more active sites, so the overpotential of 540 mV was required for NiFeP to achieve a current density of 10 mA cm^{-2} in 0.05 M H_2SO_4.

Some nonprecious metal-based acidic OER catalysts exhibit good activity or stability, but it is difficult that both can be achieved. The more active non-precious metal-based acid OER catalysts often face a rapid decay process, while the materials that can work stably for a long time are less active, which severely reduces the energy conversion efficiency. Therefore, the preparation of acid-stable and efficient nonprecious metal-based OER catalysts still faces great challenges.

3.3 | Carbon-based catalysts

3.3.1 | Heteroatoms–carbon (H–C) materials

Carbon materials have good electrical conductivity, low mass, excellent resistance to acid corrosion, and controllable architecture.[94–96] In particular, with the introduction of heteroatoms (O, N, S, P, etc.) in carbon materials, the original homogeneous sp^2 conjugation system is disrupted, and the charge density arrangement of carbon atoms adjacent to the heteroatoms is changed, which boosts their catalytic performance.[97–99] Since the potential of carbon oxidized to CO and CO_2 is much lower than the inherent potential of OER, carbon materials are susceptible to severe corrosion during the OER process.[98,100]

Some recent studies have shown that carbon materials functionalized with oxygen-containing functional groups can slow down the occurrence of carbon corrosion.[101] Moreover, the introduction of oxygen heteroatoms can activate sp^2 hybridized carbon and optimize the adsorption of reaction intermediates.[98] For example, Dai and coworkers prepared a series of multiwalled carbon nanotubes with different degrees of carboxylation by hydrothermal oxidation at different times.[102] Among them, the multiwalled carbon nanotubes with the appropriate degree of carboxylation (COOH-MWNTs) exhibited good acidic OER activity (265 mV@10 mA cm^{-2}) and could operate at a current density of 10 mA cm^{-2} for 10 h. A new mechanism was proposed to explain that COOH can improve the activity of MWNTs. The COOH-MWNT needed to provide electrons to stimulate lactone formation and hydrolysis, and then regained electrons at the end of the cycle and recover. Zhang's team prepared a carbon material compounded with phenanthrenequinone (PQ)-like moiety by introducing oxygen-containing functional groups on the surface of carbon material through a mild electro-oxidation process of graphite flakes.[103] The OER activities of different oxygen-containing groups (including hydroxyl, carboxyl, carbonyl, and epoxy groups) at the edge of graphene were evaluated using theoretical calculations. Among them, the model with two adjacent carbonyl groups (PQ-like moiety) at the edge of graphene was closest to the top of the volcano, and thus it had the lowest theoretical overpotential. Unlike other oxygen-containing models, the PQ-like part had a lower adsorption capacity for O*, thus reducing the free energy change from O* to OOH* (RDS). The acidic OER activity of this material (270 mV@10 mA cm^{-2}) was excellent due to the above reasons.

3.3.2 | Transition metals–heteroatoms–carbon (TM–H–C) materials

The introduction of atomically dispersed transition metal into H–C to construct TM–H–C systems can further tune the catalytic activity, which can be widely applied to various fields such as oxygen reduction reactions (ORR), carbon dioxide reduction reactions (CO_2RR), HER, and nitrogen reduction reactions (NRR).[104–106] Notably, it has also been reported that TM–H–C materials can effectively drive acidic OER and demonstrate good activity. TM–H–C materials can be classified into two categories: precious metal and nonprecious metal single atomic catalysis, based on the scarcity of transition metals.

Due to the unique bonding environment, precious metals doped H–C have good acidic OER activity. For example, an Ir single-atom-dispersed nitrogen-doped carbon (AD-HN-Ir) constructed by electrically driven amine induction was prepared.[107] AD-HN-Ir had unique O-hetero IrN_4 active centers, which accelerated the electron transfer from Ir atoms to other atoms and thus was more favorable for the kinetic process. AD-HN-Ir drove the acidic OER to 10 mA cm^{-2}, corresponding to an overpotential of 216 mV. Moreover, some noble metal-based TM–H–C materials are reported to compensate for the lack of stability of the bulk materials.[108] For example, the construction of Pt single atoms for acidic OER can avoid deactivation by gradual oxidation to poorly conducting platinum oxide. Liu et al.[78] constructed a $Pt_1–C_2N_2$ catalyst exhibiting good activity and stability. The strong hybridization of C/N 2p orbitals with Pt 5d orbitals resulted in a low 5d occupancy in $Pt_1–C_2N_2$. As a result, the active sites of $Pt_1–C_2N_2$ were more accessible to the reaction intermediate, which promoted the catalytic activity. Thus, the overpotential required for $Pt_1–C_2N_2$ SAC to reach 10 mA cm^{-2} was 232 mV.

On the other hand, as a cheaper alternative, nonprecious metal-based TM–H–C materials have been proposed for acidic OER. For example, Hou et al.[109] proposed that FeN_4 loaded on graphene-based carbon nanofibers can be used as an efficient catalyst for acidic OER. The FeN_4/NF/EG corresponded to a current density of 10 mA cm^{-2} at an overpotential of 294 mV. KSCN-poison test also confirmed FeN_4 as the actual active center for driving the reaction to proceed. Additionally, an acidic OER catalyst with Co single-atom sites bound to one amino- and four pyridine-N (HNC–Co) was reported.[110] The active sites in this HNC-Co were converted to $H_2N–(*O–Co)–N_4$ during the OER process, which facilitated the conversion of oxygen-containing intermediates and thus stimulated the acidic OER activity. Thus, HNC–Co can drive OER in an acidic environment with an overpotential of 265 mV at 10 mA cm^{-2}.

There are still relatively few studies on carbon materials for acidic OER applications. This is due to the fact that the stability of carbon materials is still the biggest obstacle to their application as OER catalysts.[100] Especially, when it is applied in PEMWE, the carbon material will be easily oxidized under the harsh conditions of acidic OER. Although some partially oxidized carbon materials can slow down the rapid corrosion of carbon materials to some extent, this does not fundamentally solve the long-term stability problem of carbon materials under high current densities for acidic OER. It should also be noted that the TM–H–C material was reported to be unfavorable for the OER process.[111] Although the strong coordination of heteroatoms can effectively disperse and anchor TM-single atoms, it also changes the electronic environment of metal atoms and affects the adsorption with reaction intermediates, thereby reducing the catalytic activity and efficiency.[111]

4 | ACTIVITY ENHANCEMENT STRATEGIES

The sluggish kinetic process makes acidic OER catalysts have low conversion efficiency. The promotion of acidic OER activity plays an important role in brightening the prospect of PEMWE. In this critical and promising area, many methods have been exploited to construct active acidic OER catalysts, including composition design, substrate approach, morphology strategy, atom-doping, defect engineering, strain design, and anion regulation (Figure 4).

4.1 | Composition engineering

The physicochemical properties of monometallic materials could further be adjusted by means of synergistic effects, optimized coordination environment, and electronic structure possessed by polymetallic materials.[112–115] Therefore, rational adjustment of the composition is an important method to construct highly active acidic OER catalysts.[116–118] For example, a rutile RuO_2 and CrO_2 solid solution ($Cr_{0.6}Ru_{0.4}O_2$) was prepared by Chen et al.[119] using a Cr-based MOF impregnated with Ru as a precursor by pyrolysis. The prepared catalyst required an overpotential of 178 mV to reach a current density of 10 mA cm^{-2} and had a mass activity of 229 A g^{-1} at 270 mV (Figure 5A). The introduction of Cr could adjust the electronic structure of Ru and affect the Ru–O bonding, leading to lower formation energy of *OOH which is the RDS (Figure 5B). Therefore, the construction of this bimetallic oxide significantly improved the activity of the catalyst. Besides, composition engineering is also a meaningful means to improve the activity of nonnoble metal-based catalysts. Song et al. designed Co_3O_4/CeO_2 composites with good acidic OER performance.[112] The introduction of CeO_2 improved the bonding environment and electronic structure of Co_3O_4, which directly led to the formation of Co_{IV} species with good activity on the catalyst surface as well as slowed down the charge accumulation of Co_3O_4, which significantly enhanced the catalytic activity (Figure 5C,D). The Co_3O_4/CeO_2 composite required 347 and 443 mV to reach 10 mA cm^{-2} on carbon cloth and FTO, respectively (Figure 5E). Furthermore, although the introduction of CeO_2 in Co_3O_4/CeO_2 increased the activity of the catalyst, it did not increase the Co dissolution rate, demonstrating good stability.

The perovskite oxide with the chemical formula ABO_3 is an important part of multimetal oxides.[123] Since Seitz et al.[124] pioneered the application of $SrIrO_3$ thin films in acidic OER catalysts, a large number of perovskite-type acidic OER catalysts have been reported. Most of the acidic OER catalysts revolve around perovskite oxides occupied by Ir and Ru at the B site. Notably, the adsorption capacity for reaction intermediates and surface electronic properties of the perovskite oxides are very easy to be tuned due to the high adaptability to the backbone structure and flexibility of the A and B sites (Figure 5F).[125–127] Zou et al.[120] alloyed $SrIrO_3$ (SIO) with inert $SrZrO_3$ (SZO) to construct a solid solution of perovskite oxide (SZIO) (Figure 5G). Due to the controlled catalyst size and reduced surface oxygen adsorption capacity obtained by the alloying strategy, SZIO exhibited a more pronounced activity enhancement. It reached a current density of 75 mA cm_{geo}^{-2} at 1.53 V, 15 and 3 times higher than that of IrO_2 and SIO, respectively (Figure 5H). More importantly, the configuration of the polymetallic oxide allowed for a further reduction of Ir dosage. The mass activity of SZIO

FIGURE 4 Strategies to improving the activity of catalysts

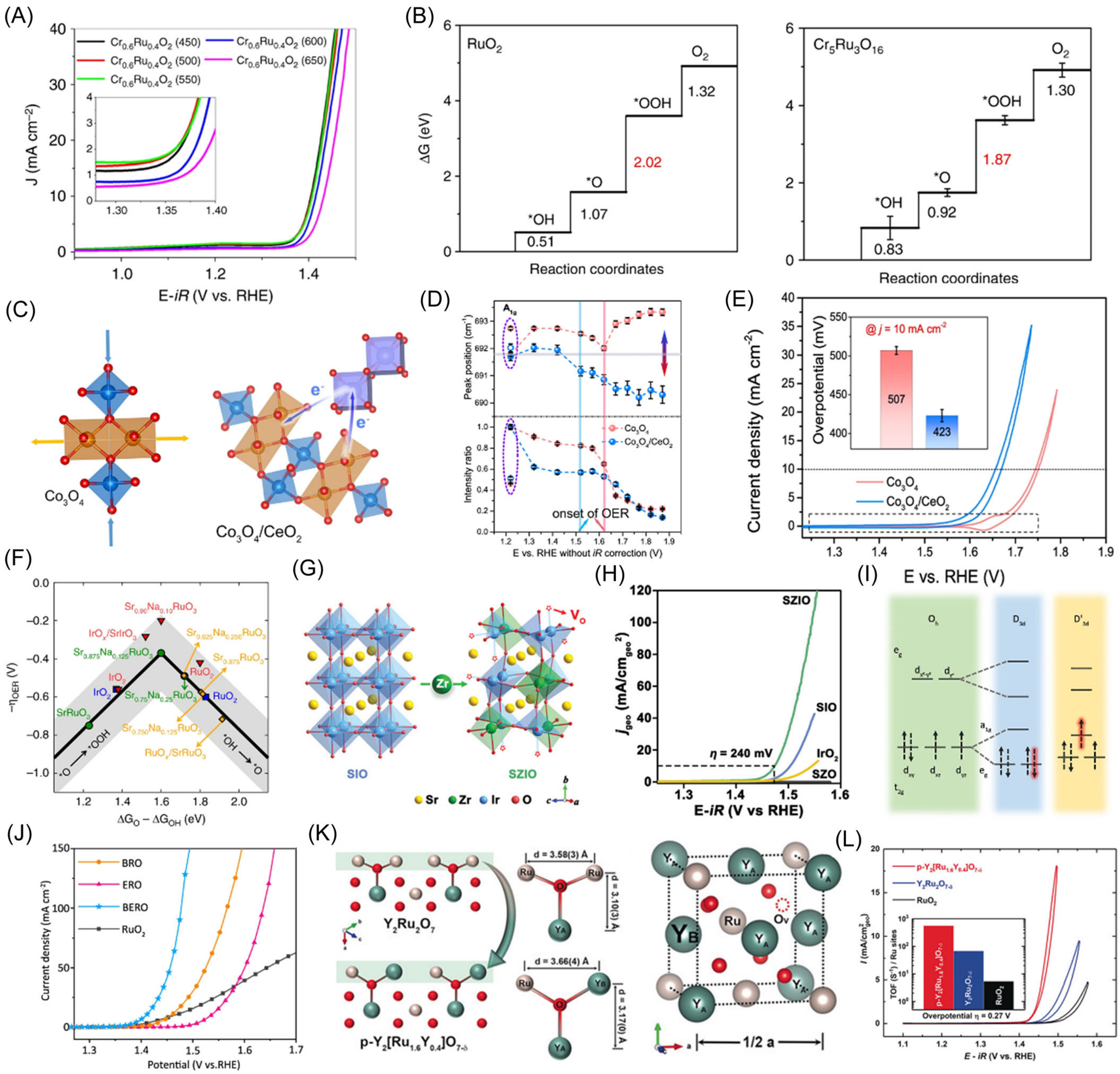

FIGURE 5 (A) LSV curves of $Cr_{0.6}Ru_{0.4}O_2$ obtained at different pyrolysis temperatures; (B) OER process at active sites on $Cr_{0.6}Ru_{0.4}O_2$ and RuO_2; Reproduced with permission.[119] Copyright 2019, Springer Nature; (C) Changes in the bonding environment of Co_3O_4 by OER and the electron transfer of Co_3O_4/CeO_2; (D) The position of the Raman A1g peak and the ratio of the corresponding intensity to the intensity at 1.22 V as a function of potential; (E) OER performance of Co_3O_4 and Co_3O_4/CeO_2; Reproduced with permission.[112] Copyright 2021, Springer Nature; (F) Volcano plots of various perovskite OER catalysts; Reproduced with permission.[24] Copyright 2019, Springer Nature; (G) Crystal structures of SIO and SZIO; (H) OER performance of SZIO and comparative samples; Reproduced with permission.[120] Copyright 2020, John Wiley and Sons; (I) Ru-4d orbital splitting for different symmetry; (J) LSV curves for OER of $Bi_xEr_{2-x}Ru_2O_7$ and control samples; Reproduced with permission.[121] Copyright 2022, Springer Nature; (K) Lattice structures of $Y_2[Ru_{1.6}Y_{0.4}]O_{7-\delta}$ and $Y_2Ru_2O_7$; (L) OER performance of $Y_2[Ru_{1.6}Y_{0.4}]O_{7-\delta}$ and control samples. Reproduced with permission.[122] Copyright 2020, John Wiley and Sons. OER, oxygen evolution reaction.

at 1.53 V (1540 A g_{Ir}^{-1}) was much higher than that of IrO_2 (39 A g_{Ir}^{-1}) and SIO (314 A g_{Ir}^{-1}).

Besides perovskites, pyrochlores with the general formula $A_2B_2O_7$ are also an important part of multimetal oxides for acidic OER. Generally, the B-sites of pyrochlore oxides are occupied by Ru and Ir and exhibit great promise for the efficient driving of acidic OER. Similar to perovskites, the highly tunable and flexible A- and B-sites allow the physicochemical properties of pyrochlore oxides to be easily engineered. For example, Liu et al.[121] modulated the A-site component to construct $Bi_xEr_{2-x}Ru_2O_7$ pyrochlores with a unique spintronic configuration. The bonding between Ru and O was affected by Bi-6s lone pair electrons, which led to the disordered atomic hybridization of $Bi_2Ru_2O_7$ at room temperature. The introduction of Er precisely adjusted the degree of atomic disorder in $Bi_xEr_{2-x}Ru_2O_7$, breaking the spin-dependent symmetry, and thus the active site changed from low spin to high spin. Therefore, the as-prepared $Bi_xEr_{2-x}Ru_2O_7$ had good OER activity and required an overpotential of about 0.18 V to reach 10 mA cm^{-2}. Yang et al.[122] designed a porous $Y_2[Ru_{1.6}Y_{0.4}]O_{7-\delta}$ pyrochlore oxide showing good acid water oxidation ability. The substitution of Y^{3+} cations for smaller Ru^{4+} cations caused lattice expansion and the production of oxygen vacancies. The absence of lattice oxygen left Ru in a mixed valence state of 4+ and 5+. Therefore, the rational design of the B site and the porous structure enabled $Y_2[Ru_{1.6}Y_{0.4}]O_{7-\delta}$ to drive a current density of 18.1 mA cm^{-2} at a voltage of 1.50 V and exhibit good acidic OER activity.

4.2 | Substrate engineering

In addition to optimizing composition, reducing the catalyst size to atom, cluster, or small particle scale and compounding it with a suitable carrier is a promising way to boost activity.[128] Supports with a high surface area loaded with active sites can enhance the accessibility of active centers.[129] Moreover, the coupling of metal particles or atoms and the carrier will generate significant electron transfer and influence the adsorption for reaction intermediates, thus improving the catalyst performance.[130,131]

Single-atom catalysts exhibit unique electronic states due to strong metal-support interactions (SMSI), quantum size effects, and unique coordination environments.[78,111,132–134] For example, the Xi group prepared Ir single atoms-loaded $NiCo_2O_4$ porous nanosheets by coelectrodeposition (Figure 6A).[135] The Ir single atoms anchored by oxygen vacancies on the $NiCo_2O_4$ porous nanosheets stimulated the OER activity and stability of the catalyst. Ir-$NiCo_2O_4$ NSs required an overpotential of 240 mV to drive 10 mA cm^{-2}, and could maintain long-term stability at this current density for 70 h (Figure 6B). The material had high mass activities of 10.00 and 28.89 A mg_{Ir}^{-1} at 1.467 and 1.508 V, respectively. The introduction of Ir single atoms promoted the adsorption and further splitting of H_2O, and also avoided the problems of overoxidation or valence instability of Co sites (Figure 6C). Ru single atoms anchored on the substrate were also confirmed to have excellent acidic OER activity. Ru–N sites supported on C_3N_4 have been reported as an excellent OER catalyst (Figure 6D).[136] This carbon nitride with a large number of electron lone pairs and unsaturated nitrogen is a suitable support for anchoring Ru single atoms. The corresponding overpotential of Ru–NC at a current density of 10 mA cm^{-2} was only 267 mV, which is lower than that of RuO_2 (300 mV) (Figure 6E). Moreover, at the voltages of 1.497 and 1.530 V, the mass activities of Ru–NCs were 3571 and 14284 A g_{Ru}^{-1}, which respectively were 322 and 410 times higher than the corresponding values of RuO_2/C.

Loading catalyst particles with controlled size on suitable substrates is also an effective means to improve activity.[139–141] For example, RuO_2 nanoparticles loaded on WC with good corrosion resistance and electrical conductivity served as good active acidic OER catalysts (Figure 6G).[137] The introduction of WC substrate made RuO_2 nanoparticles on the surface show better electron transfer ability. Moreover, the electrons of W atoms in the WC substrate would be transferred to Ru, thus making Ru exhibit an electron-rich state (Figure 6F). As a result, the electronic state of Ru was improved, which significantly optimized the adsorption of reaction intermediates. The overpotential of RuO_2-WC NPs corresponding to 10 mA cm^{-2} was 347 mV, which was slightly higher than that of RuO_2 (Figure 6H). Nevertheless, the mass activity of RuO_2-WC NPs at 10 mA cm^{-2} (1430 A g_{Ru}^{-1}) was higher than that of the substrate-free sample (176 A g_{Ru}^{-1}) by a factor of eight (Figure 6I). Besides, graphitic carbon nitride (GCN) has been shown to be a good substrate for IrO_2 nanoparticles (Figure 6L).[138] The low-coordinated IrO_2 nanoparticles were uniformly and firmly anchored on the surface of GCN nanosheets. The hydrophilicity of GCN nanosheets ensured sufficient exposure to active sites and efficient mass transfer (Figure 6J). The electron density and lattice strain near the low-coordinated Ir were optimized so that the adsorption strength for the reaction intermediates was tuned. Thus, the IrO_2/GCN had a high mass activity of 1280 mA mg^{-1} at 1.6 V, and required the overpotential of only 276 mV to drive 10 mA cm^{-2} (Figure 6K).

4.3 | Morphology engineering

The morphology is an important aspect in determining the catalyst activity. Most of the OER occurs only in the surface and near-surface regions of the catalyst.[43] Therefore, the materials with a higher surface area to volume ratio can expose more active sites and thus increase catalytic activity.[142] Moreover, catalyst structures with open

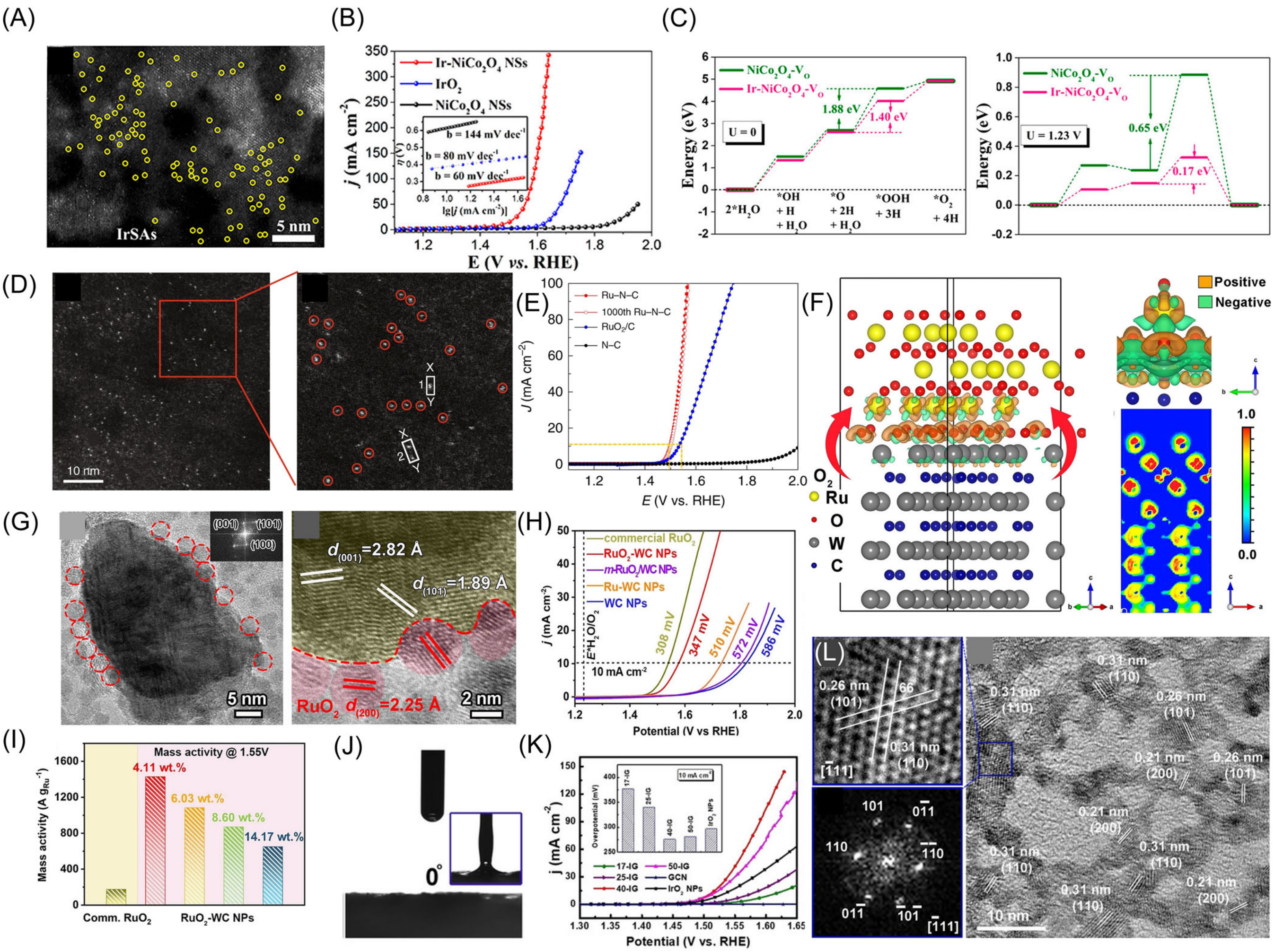

FIGURE 6 (A) High-angle annular dark-field scanning transmission electron microscope (HAADF-STEM) image of Ir–$NiCo_2O_4$; (B) LSV curves of Ir–$NiCo_2O_4$ NSs and comparison samples; (C) Gibbs free energy diagrams of OER at 0 and 1.23 V, respectively; Reproduced with permission.[135] Copyright 2020, American Chemistry Society; (D) HAADF-STEM images of Ru-NC; (E) LSV curves of Ru-NC and other samples; Reproduced with permission.[136] Copyright 2019, Springer Nature; (F) Schematic diagram of electron transfer at the RuO_2(100)-WC(001) heterointerface; (G) HRTEM image of RuO_2 dispersed on the WC surface; (H, I) Polarization curves and mass activity of RuO_2–WC NPs and control samples; Reproduced with permission.[137] Copyright 2022, John Wiley and Sons; (J) Hydrophilicity of GCN; (K) OER activity of samples such as IrO_2/GCN; (L) HRTEM image of IrO_2/GCN. Reproduced with permission.[138] Copyright 2022, John Wiley and Sons. OER, oxygen evolution reaction.

structures can enhance the material transfer efficiency.[22] Therefore, the construction of highly open morphologies is crucial to achieving enhanced activity of acidic OER catalysts. The design of acidic OER catalysts with highly open structures can be achieved mainly through the construction of low-dimensional materials and porous structures.

Low-dimensional materials have ultrahigh atomic utilization and tunable electronic properties. Therefore, constructing catalysts into low-dimensional materials will have a positive impact on the improvement of catalytic activity.[143] By adjusting the amount of cysteamine in the molten salt system to orientate the morphology, an IrO_2 with a diameter of 2 nm and a nanoneedle morphology was prepared (Figure 7A).[144] The nanoneedles were about 30 nm long and have six–eight layers (110) planes. And compared with bulk particles, nanoneedle-like IrO_2 showed better electrical conductivity. Due to the construction of this needle-like morphology, the acid OER activity of the material was significantly improved. The nanoneedle-like IrO_2 achieved a current density of 10 mA cm^{-2} at an overpotential of 0.313 V (Figure 7B). At a voltage of 1.55 V, the mass activity of nanoneedle-like IrO_2 was more than three times that of bulk IrO_2 (Figure 7C). Constructing catalysts as two-dimensional (2D) materials with high atomic utilization is also an effective means to increase the exposure of active centers. Shao et al.[145] prepared an

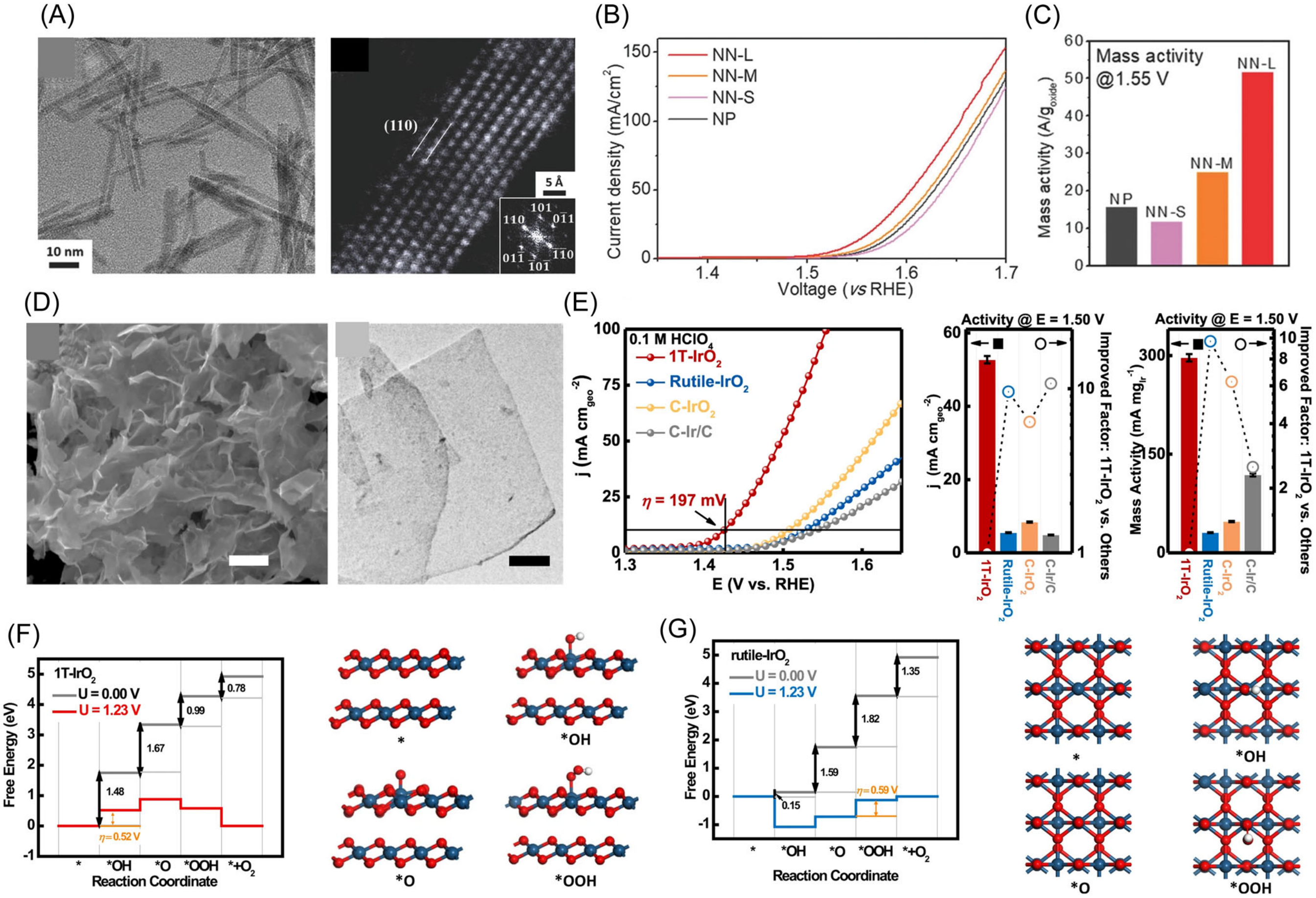

FIGURE 7 (A) Scanning electron microscope and high-angle annular dark-field scanning transmission electron microscope (HAADF-STEM) images of nanoneedle-like IrO_2; (B) Polarization curves and (C) mass activity of IrO_2 with different morphologies; Reproduced with permission.[144] Copyright 2018, John Wiley and Sons; (D) Morphology of 1T-IrO_2; (E) Acidic OER performance of 1T-IrO_2 with other samples; (F, G) Free energy change and schematic diagram of OER process on (F) 1T-IrO_2 and (G) rutile IrO_2. Reproduced with permission.[145] Copyright 2022, Springer Nature.

ultrathin 2D layered 1T phase-iridium oxide by mechanochemical synthesis in a strong alkaline and high-temperature environment (Figure 7D). The construction of this 2D morphology allowed more Ir atoms to participate in the reaction and effectively promoted mass transfer. Additionally, unlike the relatively exposed Ir atoms in rutile IrO_2, the Ir atoms in this 1T phase IrO_2 with a 2D structure were buried by O atoms, which optimized the adsorption of hydroxyl groups on the active sites (Figure 7F,G). This 1T-IrO_2 required an overpotential of 197 mV to reach 10 mA cm^{-2}, and had a mass activity at 1.5 V (296.8 mA mg_{Ir}^{-1}) which was about 9.7 times that of rutile IrO_2 (Figure 7E).

Porous materials expose a large number of catalyst–electrolyte interfaces, resulting in more active sites driving reactions.[143] The main strategies for constructing porous structures include the template method and selective etching strategy. The use of templates as structural guides to control the catalyst morphology in a directional manner allows the construction of catalyst materials with more exposed active sites and lower diffusion difficulty (Figure 8A). The amphiphilic triblock copolymer (PEO–PB–PEO) was used as a soft template to prepare nanocrystalline iridium oxide films with ordered pores of about 16 nm in diameter.[146] The surface area of the mesoporous IrO_2 films constructed by template mediation was 2.1 times higher than that of the untemplated material (Figure 8B). The activity of templated IrO_2 had a significant enhancement compared to the untemplated sample. The voltage required to achieve a current density of 100 mA cm^{-2} for templated IrO_2 was lower than that of untemplated IrO_2 by about 40 mV (Figure 8C). The voltammetric curves normalized by the active surface area were not significantly different, indicating that templating method did not change the intrinsic activity of the surface IrO_2, but only increased the degree of exposure (Figure 8D).

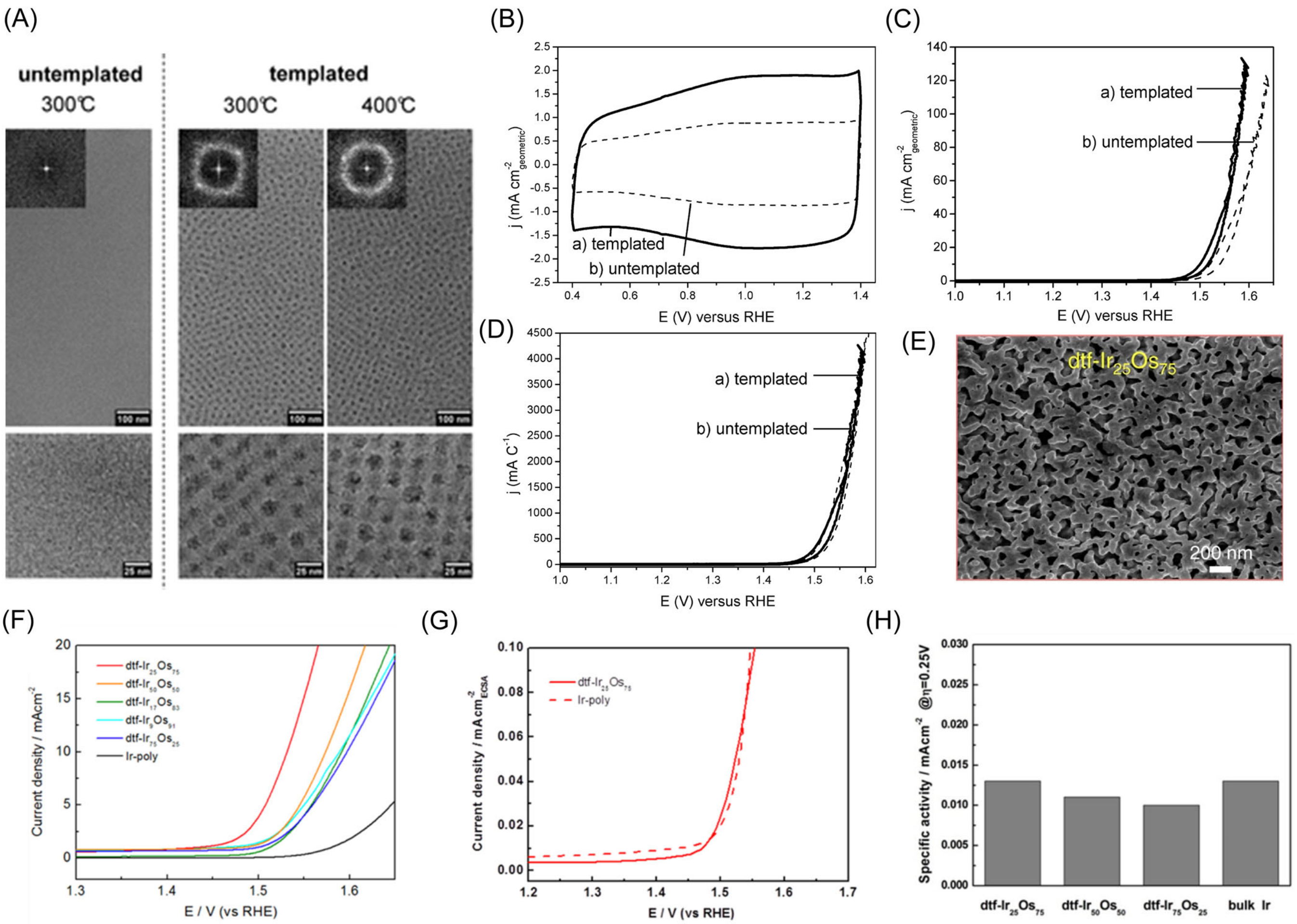

FIGURE 8 (A) SEM images of templated and untemplated samples obtained by pyrolysis at different temperatures; (B, C) Cyclic voltammetry (CV) curves of templated and untemplated IrO_2; (D) ECSA-normalized CV curves; Reproduced with permission.[146] Copyright 2011, American Chemistry Society; (E) SEM image of dtf-$Ir_{25}Os_{75}$; (F) Polarization curves of dtf-$Ir_{25}Os_{75}$ and other samples; (G) LSV curves of IrOs alloys normalized by ECSA before and after dealloying; (H) Specific activities of IrOs alloys with different Os proportions after dealloying. Reproduced with permission.[147] Copyright 2017, Springer Nature. ECSA, electrochemically active surface area; SEM, scanning electron microscope.

Selective etching to construct structures with more openness is also an effective way to increase the number of exposed active sites and enhance mass activity. Li et al.[148] electrochemically leached Sr from bulk $SrRuO_3$ ceramics to construct highly exposed RuO_2 clusters accompanied by lattice collapse. Such RuO_2 clusters were produced with significantly increased electrochemically active surface area (ECSA) and more exposed active sites. Moreover, the electrochemical leaching was accompanied by a more pronounced Jahn–Teller distortion in RuO_2, which significantly enhanced the intrinsic activity. As a result, this selectively etched sample exhibited a high mass activity of 50 mA mg^{-1} at an overpotential of 190 mV. Selective etching using the different solubility between the components of alloy materials can also achieve the construction of porous materials. For example, a porous Ir/IrO_2 nanomaterial (dtf-$Ir_{25}Os_{75}$) was constructed from $Ir_{25}Os_{75}$ alloy after rapid dealloying of Os (Figure 8E).[147] The ECSA of $Ir_{25}Os_{75}$ after dealloying was about 50 times higher than that of Ir nanoparticles. This indicated that the removal of Os exposed more surface area and led to a higher surface area-to-volume ratio, which enhanced the availability of active sites. The dtf-$Ir_{25}Os_{75}$ exhibited superior activity than Ir nanoparticles, but they had similar activity after ECSA-normalization (Figure 8F–H). This demonstrated the significant modulation of the catalytic activity by the morphology design.

4.4 | Atom-doping engineering

In the case of the OER process, bond formation or breakage depends on the oxygen-containing intermediates bound to active sites.[149–152] Therefore, the activity of the catalyst is determined by its d-orbital electronic structure.[151,152] Notably, doping with exotic elements can significantly improve the d-orbital state and thus the excitation of the catalytic activity.[153] The two main types of doping of foreign elements are cationic and anionic doping.

Cationic doping for improving the intrinsic activity of Ru and Ir-based oxides has been studied extensively. A large number of metal elements (Ni, Co, Cr, Fe, Ce, Cu, Mn, V, etc.) have been shown to have good improvement in the acidic OER activity of the pristine materials.[154–160] Chen et al.[160] produced an Mn-doped RuO_2 material with a hollow structure by replacing Mn in the Mn-based metal-organic framework (Mn-BTC) with Ru^{3+} followed by pyrolysis in air. The doped Mn made a larger difference between the d-band center of the Ru site and the p-band center of the adsorbed oxygen, which reduced the RDS free energy (Figure 9B,C). Thus, the performance of RuO_2 was significantly enhanced by Mn doping, with overpotentials of 158 mV required to drive 10 mA cm^{-2} for Mn–RuO_2 (Figure 9A). Additionally, metal atom doping can break the scaling relationship of the reaction intermediates, thereby achieving better OER activity.[161] For example, the doping of Ni and Co at the surface bridging sites could activate the oxygen atoms on top of them that do not otherwise participate in the OER process (Figure 9D).[161,166] These oxygen atoms acted as proton donor-acceptors to make *OH and *OOH more susceptible to deprotonation, thereby reducing the free energy required for the reaction.[166]

It has also been reported that the doping of alkaline-earth metals can construct amorphous iridium oxide with higher activity by changing the long-range ordering of the lattice (Figure 9E,F,H).[63] Theoretical calculation showed that the doping of Li will make the Li–IrO_x system thermodynamically more OER-friendly in terms of electronic structure. Moreover, unlike the hindered kinetics of hydroxyl oxidation and O–O bond formation on rutile IrO_2, the lattice distortion, and amorphous structure due to doped Li would accelerate the progress of these two steps, thus having more favorable kinetics of the OER process (Figure 9G). Additionally, the doping of metal atoms can affect the reconfiguration process of the catalyst surface. For example, Fu et al.[162] showed that V doping led to the transformation of the surface of CoP_2 into an oxygen vacancy-rich Co_3O_4 layer under acidic OER conditions, which modulated the adsorption and desorption processes of intermediates and consequently reduced the RDS potential (Figure 9I). Thus, the V-CoP_2/CC required an overpotential of only 320 mV to achieve 50 mA cm^{-2} compared to CoP_2/CC (561 mV@50 mA cm^{-2}) and RuO_2/CC (497 mV@50 mA cm^{-2}), proving its better acidic OER activity (Figure 9J).

On the other hand, some studies have also pointed out that the doping of anions also significantly affects the electronic state of the active site, thus enabling the modulation of the adsorption capacity for the OER intermediates.[167] For example, Kumta's team prepared a series of IrO_2 with different F doping amounts by a wet chemistry method, and the results showed that F doping can effectively enhance the OER activity of IrO_2 in acidic environments.[163] The samples containing 20 wt.% F exhibited the best performance compared to other F-doped samples, and F doping significantly induced p, d hybridization between the Ir 5d and F 2p states, thus reducing the free energy of *OOH formation (RDS) (Figure 9K,L).[164] Besides, S doping has been shown to significantly improve the catalytic activity of $SrIrO_3$.[165] The mass activities of undoped M–$SrIrO_3$ and commercial IrO_2 at 1.55 V were 12.3 and 4.3 A g_{Ir}^{-1}, respectively, both much lower than that of S-doped M-$SrIrO_3$ (41.5 A g_{Ir}^{-1}) (Figure 9M). The DOS curve of S-doped M–$SrIrO_3$ crossed the Fermi energy level and had a narrower band gap than that of the undoped sample, which demonstrated that the S doping increases the electronic conductivity of the original sample (Figure 9N). Moreover, doped S improved the strong binding of oxygen-containing intermediates on the catalyst surface (Figure 9O). The Ir on the surface of S-doped M-$SrIrO_3$ had a lower oxidation state, which was an indication of the lower adsorption free energy of the material. At present, there are relatively few studies on anion doping to improve the activity of acidic OER, and exploration of this aspect is critical and necessary.

4.5 | Defect engineering

Defects can significantly affect the physicochemical properties of materials and have become one of the important parts of the field of catalysis.[151,168–170] The introduction of appropriate defects can improve the surface chemical state of the active site and regulate the over- or under-adsorption of the active site to the reaction intermediates, thus achieving a significant excitation of the catalytic activity.[171,172] The basic strategies for constructing defects according to the stages of defect formation include in situ synthesis and postprocessing methods.

In situ synthesis is a means of introducing defects directly into the material during the synthesis process. For example, a layered RuO_2 with complex Ru vacancy defects (RuO_2 NSs) prepared using the molten salt method was reported (Figure 10C).[173] Such 2D lamellar RuO_2 NSs showed significantly enhanced catalytic activity, corresponding to an overpotential of 199 mV

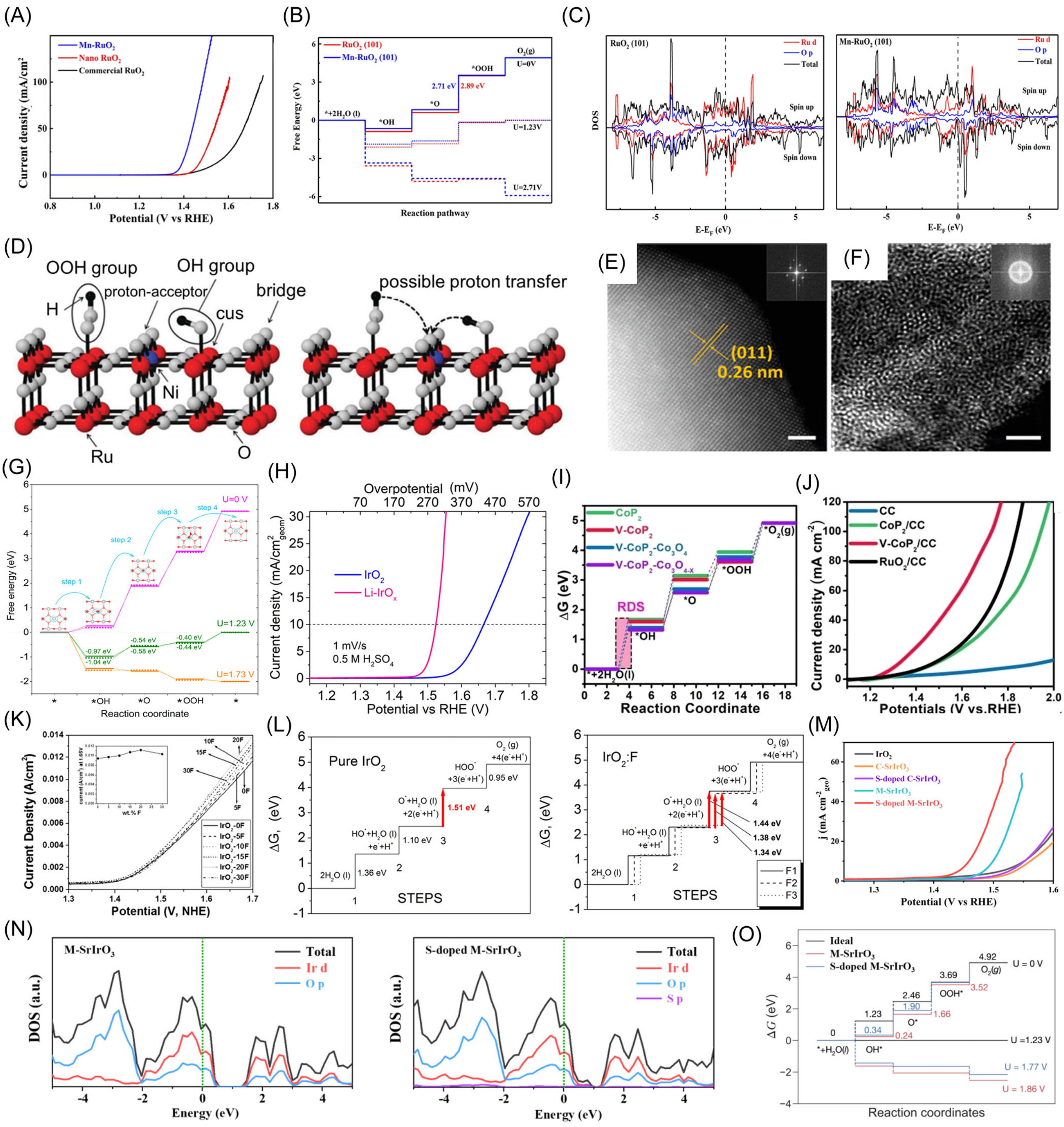

FIGURE 9 (A) Polarization curves for OER of Mn–RuO_2 and comparative samples; (B) Changes in the free energy of RuO_2 before and after Mn doping; (C) The DOS curves of RuO_2 and Mn–RuO_2; Reproduced with permission.[160] Copyright 2019, American Chemistry Society; (D) Effects of Ni and Co doping on proton transfer; Reproduced with permission.[161] Copyright 2017, John Wiley and Sons; (E, F) HAADF–STEM images of rutile-IrO_2 and amorphous IrO_x; (G) The change in free energy as the reaction proceeds; (H) OER polarization curves of IrO_2 and amorphous Li-IrO_x; Reproduced with permission.[63] Copyright 2019, American Chemistry Society; (I) Gibbs free energy changes for acid OER of V-CoP_2 and other samples; (J) LSV curves of V-CoP_2/CC and control samples; Reproduced with permission.[162] Copyright 2022, John Wiley and Sons; (K) LSV curves of F-doped IrO_2; Reproduced with permission.[163] Copyright 2013, Elsevier, Inc; (L) The effect of F doping on the adsorption energy of IrO_2; Reproduced with permission.[164] Copyright 2013, American Chemistry Society; (M) The performance of S-doped M-$SrIrO_3$ and the comparison sample; (N, O) The DOS and free energy changes for OER of S-doped M-$SrIrO_3$ and M-$SrIrO_3$. Reproduced with permission.[165] Copyright 2021, Elsevier, Inc.

at the current density of 10 mA cm_{geo}^{-2}, which was significantly higher than that of commercial RuO_2 ($\eta = 325$ mV) and annealed RuO_2 NSs ($\eta = 244$ mV) (Figure 10A). Moreover, at an overpotential of 230 mV, the RuO_2 NSs had significantly enhanced mass activity (0.52 A mg_{Ru}^{-1}) and specific activity (0.89 mA cm_{oxide}^{-2}), which were 80.6 and 14.9 times higher than those of commercial RuO_2 (Figure 10B). The DFT results showed that the reaction process of defect-free RuO_2 was hindered by the OOH* formation and lay at the bottom of the 2D volcano bottom of the diagram (Figure 10D). The introduction of Ru vacancy defects modulated the energy in the conversion from O* to OOH*, resulting in a catalytic activity closer to the top of the volcano. The introduction of oxygen vacancies has also been shown to modulate the acidic OER activity of the catalysts. Using graphene oxide (GO) as an oxygen source for limited Ru oxidation, Wu et al.[174] prepared a GO-based RuO_2

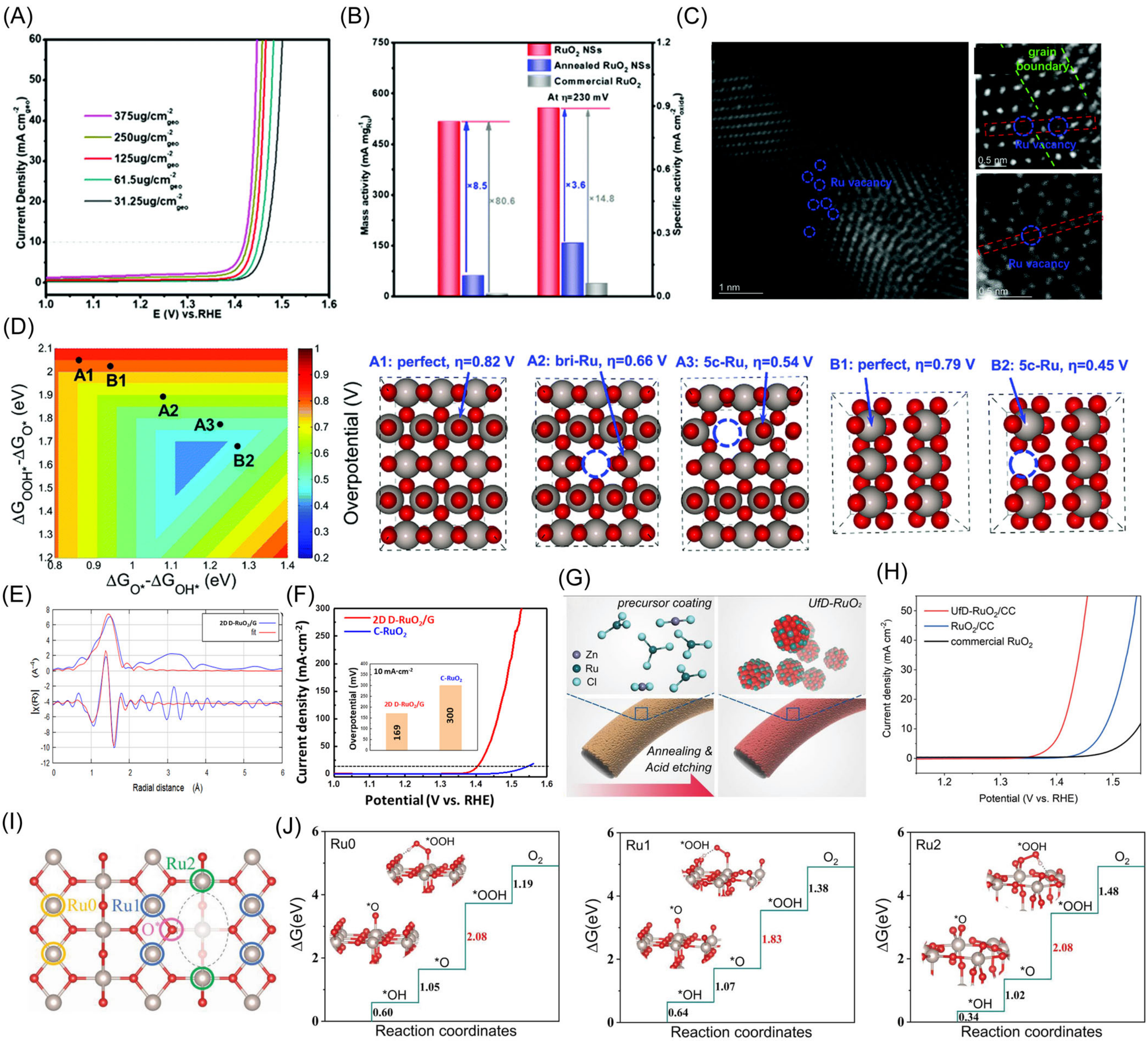

FIGURE 10 (A) LSV curves and (B) mass and specific activity of defective, annealed, and commercial RuO_2; (C) HR-STEM images of RuO_2 NSs; (D) Volcano plot based on $\Delta G_{O*}-\Delta G_{OH*}$ and $\Delta G_{OOH*}-\Delta G_{O*}$; Reproduced with permission.[173] Copyright 2020, Royal Society of Chemistry; (E) EXAFS k space fitting and FT-EXAFS fitting curve of Ru K-edge in 2D D-RuO_2/G; (F) LSV curves for OER of 2D D-RuO_2/G and C-RuO_2. Reproduced with permission.[174] Copyright 2020, Elsevier, Inc; (G) UfD-RuO_2/CC prepared by acid elution of Zn; (H) Performance of UfD-RuO_2/CC with other samples; (I, J) (I) Models of different types of Ru sites and (J) their free energy changes for OER. Reproduced with permission.[175] Copyright 2019, John Wiley and Sons.

with unsaturated Ru–O coordination (2D D-RuO_2/G) (Figure 10E). The oxygen-vacancy-rich structure allowed the Ru site to have a locally asymmetric coordination environment, which significantly reduced the RDS potential from *O to *OOH significantly. Moreover, this oxygen-vacancy-rich structure allowed Ru to have a more positive state, thus enhancing the oxidation capacity. Thus, 2D D-RuO_2/G had a better acidic OER activity (169 mV@10 mA cm^{-2}), which was significantly better than 2D-RuO_2 (289 mV@10 mA cm^{-2}) and C-RuO_2 (300 mV@10 mA cm^{-2}) (Figure 10F).

The method of processing the precursors to construct defects is the post-processing method. By doping more soluble atoms into the catalyst as precursors, materials rich in atomic vacancies can be prepared by subsequent etching. For example, Zn-doped RuO_2 could be fabricated into RuO_2 with a large number of fine defects (UfD-RuO_2/CC) after etching away soluble Zn in a strong acid environment (Figure 10G).[175] The RuO_2 had more unsaturated Ru sites as active centers after constructing a large number of oxygen vacancy defects. Moreover, the introduction of defects led to a change in the electronic structure of the original five-ligand active site, which speeded up the RDS (Figure 10I,J). Therefore, UfD-RuO_2/CC had a significantly enhanced catalytic activity. In 0.5 M H_2SO_4, UfD-RuO_2/CC required an overpotential of only 179 mV to reach a current density of 10 mA cm^{-2} (Figure 10H). UfD-RuO_2/CC had 27.2 and 20.7 times higher current densities than RuO_2/CC at 1.43 and 1.45 V, respectively.

4.6 | Strain engineering

Lattice strain is a phenomenon in which the surface or partial region of a particle has a different atom-to-atom distance than the bulk material.[176,177] The strategy to adjust the electronic structure of catalysts by directionally designing lattice strains to tune the adsorption of reaction intermediates is called strain engineering.[178,179] The applications of strain engineering in acidic OER catalysts mainly include two types: core-shell structure and construction of grain boundaries.

The lattice mismatch between the outer shell and the inner core of the core-shell structure tends to exhibit unique strain effects. For example, a PdCu/Ir nanocrystal with a core-shell structure that exhibited unique strain effects was reported (Figure 11A,B).[180] Due to the lattice mismatch with the PdCu core, an Ir shell with only four monolayer thick generated a compressive strain of 3.60% and could exclude the effect of ligand effects. Under this compressive strain, the adsorption of PdCu/Ir to oxygen-containing intermediates was weakened and had a lower d-band center, and thus exhibited a better acidic OER activity (Figure 11D,E). The PdCu/Ir nanocrystal with unique strain effects could drive 10 mA cm^{-2} at the overpotential of 283 mV and had a mass activity of 1.83 A mg_{Ir}^{-1} at the overpotential of 280 mV (Figure 11C). Moreover, IrCo@IrO_x was fabricated using IrCo nanodendrites as precursors by constructing different thick of IrO_x on their surfaces using electrochemical oxidation (Figure 11F–H).[181] As the number of IrO_x layers as the shell layer increased, the compressive strain of the IrO_x layers gradually decreased. The adsorption capacity of HOO* with the catalyst gradually increased as the strain degree became higher, so RDS shifted from the forming of HOO* to the dissociation of HOO*. Therefore, the activity of catalysts with IrO_x layers of different strain levels showed a volcano-type trend (Figure 11I,J). The material with the most proper compressive strain (1.51%) obtained by precise adjustment of the number of IrO_x layers required an overpotential of 247 mV to drive 10 mA cm^{-2} (Figure 11K).

The construction of grain boundaries is also an efficient measure for generating and retaining strain. Ru nanoparticles enriched with grain boundaries (l-Ru) were prepared by liquid laser etching (Figure 12A).[182] The grain boundaries in l-Ru resulted in lower Ru-Ru coordination and shorter bond distances than annealed Ru particles, suggesting that the grain boundary-rich Ru nanoparticles generated compressive strain (Figure 12B). The compressive strain caused a change in the surface electron density of Ru atoms in the ruthenium nanoparticles and weakened the over-adsorption of active sites to oxygen-containing intermediates (Figure 12C). This grain boundary-rich ruthenium nanoparticle had good catalytic activity under the effect of compressive strain and required the overpotential of 202 mV to drive 10 mA cm^{-2}, which was much lower than that of the annealed Ru particles (284 mV) (Figure 12D). The abundance of fine grain boundaries leads to the occurrence of lattice torsion, which in turn changes the properties of the material. For example, a $Ta_xTm_yIr_{1-x-y}O_{2-\delta}$ crystal with abundant fine grain boundaries was prepared by rapid heating and pyrolysis (Figure 12E).[183] The rapid pyrolysis process led to the formation of plentiful nuclei in the precursor, and the subsequent unique mesocrystalline growth produced many twins, dislocations, and stacking layer dislocations at the fine grain boundaries (Figure 12F,G,I). To further match the crystallographic orientation, this would lead to the occurrence of lattice torsion. Under the effect of torsional strain, GB-$Ta_{0.1}Tm_{0.1}Ir_{0.8}O_{2-\delta}$ exhibited a more suitable Ir-O bond length and had a proper binding ability to oxygen intermediates, thus reducing the energy barrier of RDS (Figure 12J,K). The GB-$Ta_{0.1}Tm_{0.1}Ir_{0.8}O_{2-\delta}$ with

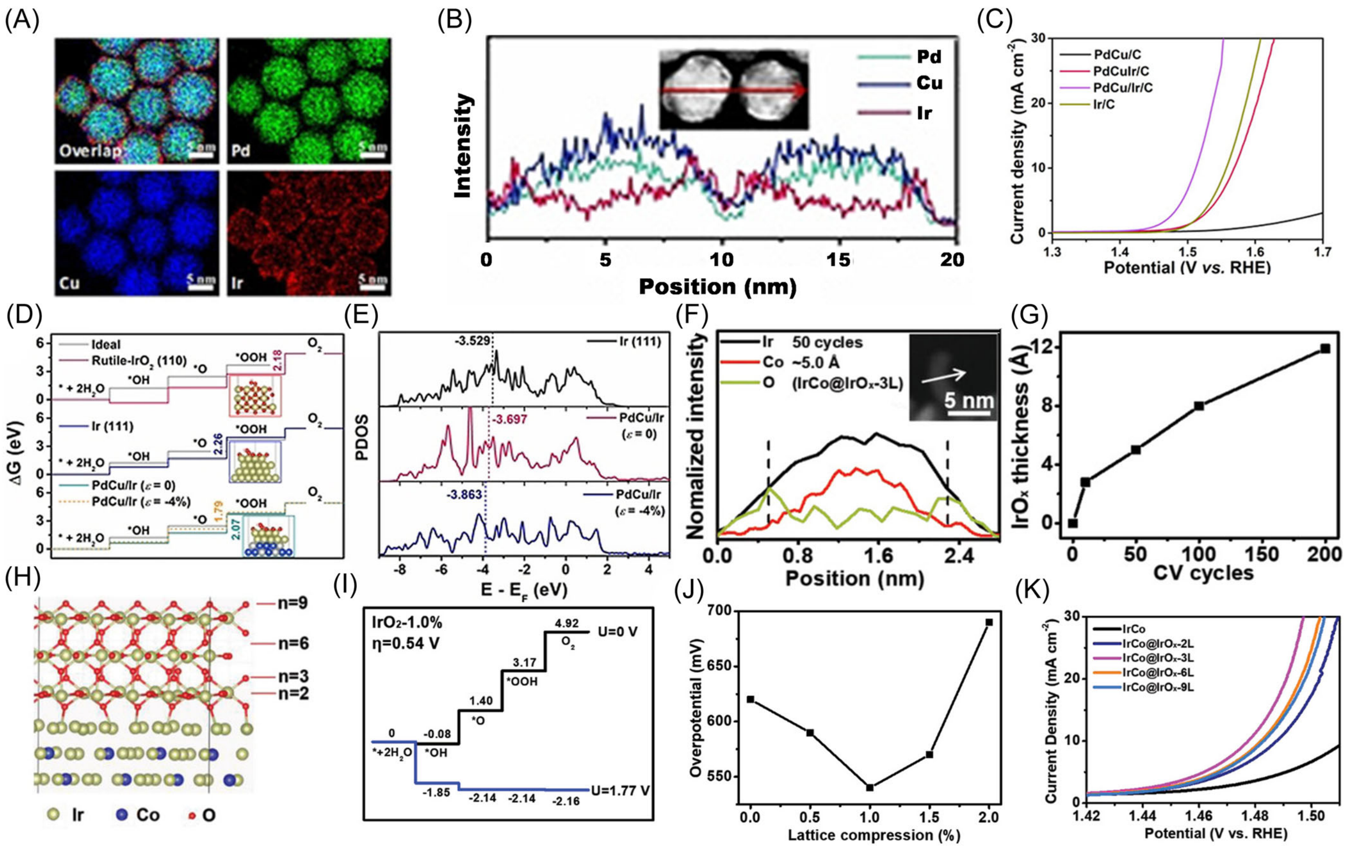

FIGURE 11 (A, B) Elemental distribution and line-scanning profiles of the PdCu/Ir core/shell structure; (C) OER performance of PdCu/Ir/C and other samples; (D) Free energy changes for OER of different electrocatalysts; (E) DOS curves of PdCu/Ir and Ir; Reproduced with permission.[180] Copyright 2021, John Wiley and Sons. (F) Line-scanning profiles of IrCo@IrO$_x$; (G) Variation of IrO$_x$ thickness with the number of cycles; (H) Model of IrCo@IrO$_x$; (I) Free energy diagram of IrCo@IrO$_x$ with the most suitable compression strain; (J) Variation of the theoretical overpotential with the degree of lattice compression; (K) OER activity of samples with different IrO$_x$ thicknesses. Reproduced with permission.[181] Copyright 2019, John Wiley and Sons. OER, oxygen evolution reaction.

the best performance required a voltage of 1.766 V to achieve the current density of 1 A cm^{-2} and had a TOF of 2.54 O_2 s^{-1} corresponding to the current density of 1.5 A cm^{-2} (Figure 12H).

4.7 | Anion regulation engineering

Anion modulation plays a positive role in the OER activity of the material.[29,184] The introduction of anions to build compounds can further adjust the local electron redistribution of metal atoms and properly adjust their adsorption strength to reaction intermediates.[185–188] For example, a pure phase RuB_2 was prepared in an environment where KCl and LiCl were salt melts (Figure 13A).[189] RuB_2 could drive a current density of 10 mA cm^{-2} at an overpotential of 223 mV, exhibiting an activity that far exceeds that of RuO_2 (294 mV@10 mA cm^{-2}). Theoretical calculations showed that the introduction of B adjusted the adsorption capacity of the active sites for the reaction intermediates, which reduced the RDS free energy from O* to *OOH (Figure 13B). Huang et al.[190] constructed an amorphous $RuTe_2$ nanorod with distorted Ru–Te bonds. The Te site had significant p–π electron-rich properties, and thus it was very electron-sensitive to coupling O–2p orbitals to activate H_2O (Figure 13C,D). The distortional strain effect of the amorphous structure enhanced the p–d transfer and thus made the p–π coupling more sensitive (Figure 13E,F). Thus, the distorted Ru–Te bonding eliminated the crystal-field-splitting effect of the Ru site and thus optimized the electronic activity near the Fermi energy level (Figure 13G,H). As a result, this amorphous $RuTe_2$ showed good acidic OER activity and required an overpotential of 245 mV to achieve 10 mA cm^{-2}.

In addition, the construction of heterogeneous structures by introducing anions has a beneficial effect on enhancing catalytic activity.[187] Electron transfer at the heterointerface affects the electron distribution,

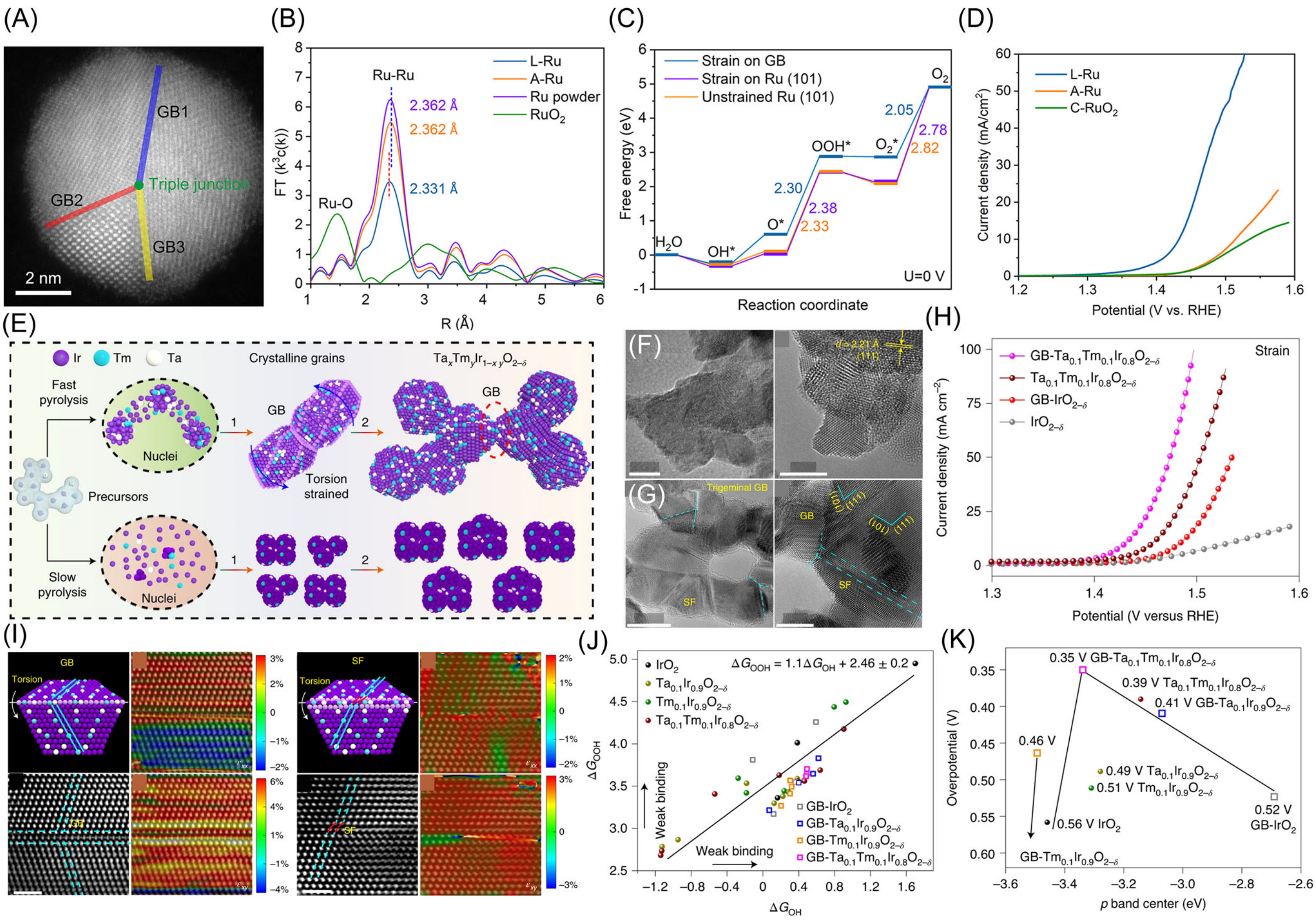

FIGURE 12 (A) The grain boundary-rich structure of L-Ru; (B) FT-EXAFS spectra of L-Ru and other samples; (C) The effect of compressive strain on the free energy of reaction; (D) LSV curves for OER of L-Ru and other samples; Reproduced with permission.[182] Copyright 2020, American Chemical Society; (E) The generation process of $Ta_xTm_yIr_{1-x-y}O_{2-\delta}$; (F, G) HRTEM images of $Ta_{0.1}Tm_{0.1}Ir_{0.8}O_{2-\delta}$ with and without fine grain boundaries; (H) The contribution of fine grain boundaries to the properties of $Ta_{0.1}Tm_{0.1}Ir_{0.8}O_{2-\delta}$ and $IrO_{2-\delta}$; (I) Geometric phase analysis of grain boundaries and stacking fault; (J)Relationship between ΔG_{OH} and ΔG_{OOH}; (K) The performance volcano plots of various $Ta_xTm_yIr_{1-x-y}O_{2-\delta}$ catalysts. Reproduced with permission.[183] Copyright 2021, Springer Nature. OER, oxygen evolution reaction.

thereby changing the electronic structure of the active site.[187] And the abundant heterointerface also exposes more edges with good catalytic activity, thus providing more active sites.[193] Our group has constructed a Ru/RuS_2 heterostructure by ionothermal method using K_2S as both reducing agent and sulfur source (Figure 13I).[191] Theoretical calculation showed that the Ru/RuS_2 heterostructure had a high state density of Fermi energy level states and thus good electronic conductivity (Figure 13K). Moreover, the charge density near the interface was redistributed and electrons were transferred from the surface of Ru to the S atom (Figure 13J). As a result, the adsorption ability of Ru atoms lacking electrons on the surface to the reaction intermediates was tuned, thus reducing the thermodynamic energy barrier (Figure 13L). For the above reasons, Ru/RuS_2 for OER catalysts exhibited relatively good catalytic activity (201 mV@10 mA cm^{-2}). Recently, the construction of oxyhalide by introducing halogen atoms has also been shown to be a way to significantly enhance catalytic activity. A manganese oxybromide ($Mn_{7.5}O_{10}Br_3$) material with good activity and stability was constructed by Zhang et al.[192] $Mn_{7.5}O_{10}Br_3$ exhibited significantly increased activity (295 ± 5 mV@ 10 mA cm^{-2}) compared to MnO_2 (413 ± 5 mV@10 mA cm^{-2}). The autoxidation process allowed the material to produce a tightly packed oxide surface during the OER process, which led to lower adsorbate-metal coordination thus further reducing the adsorption of active sites to the OER intermediates (Figure 13M–O). Moreover, the Mn–halogen interactions significantly promoted the electron transport ability of the catalyst. Thus, this manganese oxybromide material had significantly enhanced acidic OER activity.

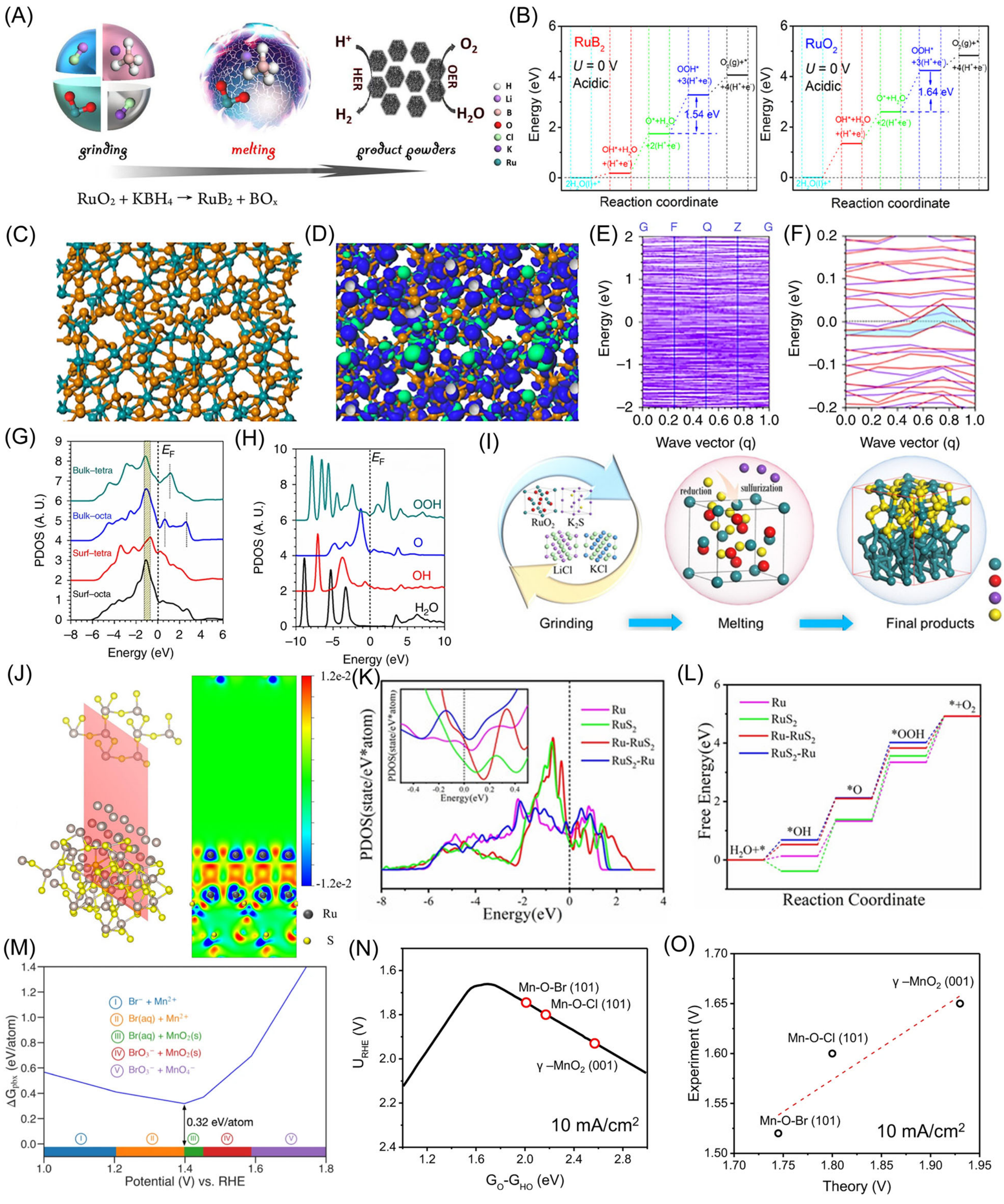

FIGURE 13 (A) Synthesis of the RuB_2; (B) Free energy changes for OER of RuB_2 and RuO_2. Reproduced with permission.[189] Copyright 2020, American Chemical Society; (C) Model of amorphous $RuTe_2$; (D) Schematic illustration of bonding and antibonding orbitals near the Fermi level; (E) Trends in structure formation energy related to strain effects; (F) Comparison of f dielectric functions; (G) DOS changes in the Ru-4d band; (H) DOS contribution of O species during OER; Reproduced with permission.[190] Copyright 2019, Springer Nature; (I) Preparation process of Ru/RuS_2; (J) The charge density difference of Ru/RuS_2; (K) DOS curves of Ru/RuS_2 and other samples; (L) Gibbs free energy changes of Ru/RuS_2 and other samples; Reproduced with permission.[191] Copyright 2021, John Wiley and Sons; (M) Pourbaix decomposition free energy of $Mn_{7.5}O_{10}Br_3$ at pH 0; (N) Volcano plot at 10 mA cm^{-2}; (O) Theoretical and actual potential at 10 mA cm^{-2}. Reproduced with permission.[192] Copyright 2019, Springer Nature. OER, oxygen evolution reaction.

5 | STABILIZATION STRATEGIES OF CATALYSTS

Stability is another important aspect of catalyst performance. The stability problem of OER catalysts is particularly bad in acidic environments. Poor stability will make the cost issue of the catalyst more prominent, and the catalyst that can work for a long time will greatly reduce the cost of the catalyst.[194] Generally, the preparation of acidic OER catalysts with high activity and stability still faces many challenges. Fortunately, many researchers have tried to delay or avoid catalyst decay to achieve the long-term operation of acidic OER catalysts. Strategies to enhance the stability of acidic OER catalysts mainly include wrapping strategy, enhancing electronic coupling, stabilizing lattice oxygen, electronic structure regulation, and self-healing (Figure 14).

5.1 | Wrapping route

The unstable inner core can be prevented from rapidly oxidizing and dissolving by constructing the outer layer cladding with better stability. Iridium-based oxides are considered ideal candidates for shells due to their better activity and stability compared to other metals.[195] Moreover, the utilization of Ir and its oxides as the shell does not increase the cost too much. For example, an Ir-coated Pd nanocube with good acidic OER activity was reported (Figure 15A).[196] The catalytic activity of Pb nanocubes covered with Ir shells with different thicknesses exhibited a volcano-type trend due to the ligand and strain phenomena caused by the heterogeneous interface of Pd and Ir (Figure 15B). Importantly, the stability of the catalyst showed a positive correlation with the number of atomic layers of the Ir shell (Figure 15C). Coatings with thicker Ir layers would contain more Ir^{4+} after electrochemical oxidation, which led to better stability (Figure 15D). The stable Ir shell layer could effectively avoid excessive oxidation of the Pd core, thereby preventing the destruction of the catalyst structure.

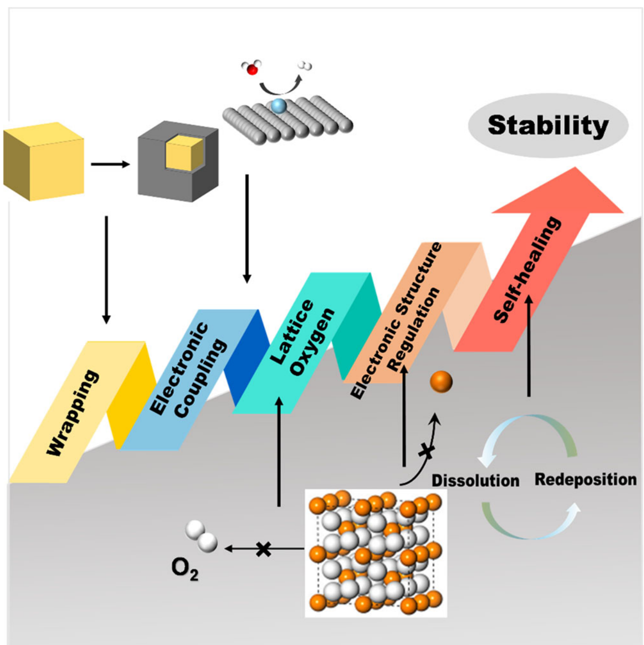

FIGURE 14 Strategies to improving the stability of catalysts

Ru-based materials have relatively high acidic OER activity but suffer from attenuation problems. Therefore, it is also promising to coat the surface of Ru-based materials with a more stable substance to improve the durability of catalysts. Qiao et al.[197] used pregenerated Ru nanocrystals to mediate the growth of Ir, and fabricated a Ru core structure covered with a thin partially oxidized Ir layer (Figure 15E). The partially oxidized Ir shell on the catalyst surface protected the Ru core from being over-oxidized, thus showing a lower chemical state. Therefore, only less Ru dissolved under the protection of the outer IrO_x shell, and the stability of the material was obviously improved (Figure 15F,G). In addition, a submonolayer IrO_x-covered RuO_2 with significantly improved stability was reported.[198] As the thickness of IrO_x deposited on the RuO_2 surface increased, the catalytic activity decreased continuously (Figure 15H). Nevertheless, due to the high dissolution potential of surface Ir, even the introduction of 1 Å thick IrO_x could significantly enhance the stability of RuO_2 (Figure 15I). The 2 Å-thick IrO_x coverage greatly reduced the metal dissolution of the catalyst during acidic OER, thus manifesting a significantly improved stability (Figure 15J,K).

5.2 | Electronic coupling route

A key to the long-term operation of acidic OER catalysts is the rational choice of substrates. The active sites on the surface of some substrates may be severely exfoliated during the OER process, resulting in performance degradation.[199] Therefore, an important aspect of the suitable substrate is that they can undergo strong electronic coupling with supported noble metals to avoid deactivation due to shedding of active species.

Carbon materials have a large specific surface area, low cost, and good electrical conductivity, so they are widely used as substrate materials.[200,201] For example, N-doped graphene sheets can achieve effective anchoring of Ir nanoparticles on their surfaces.[202] The theoretical calculation results showed that the introduction of N

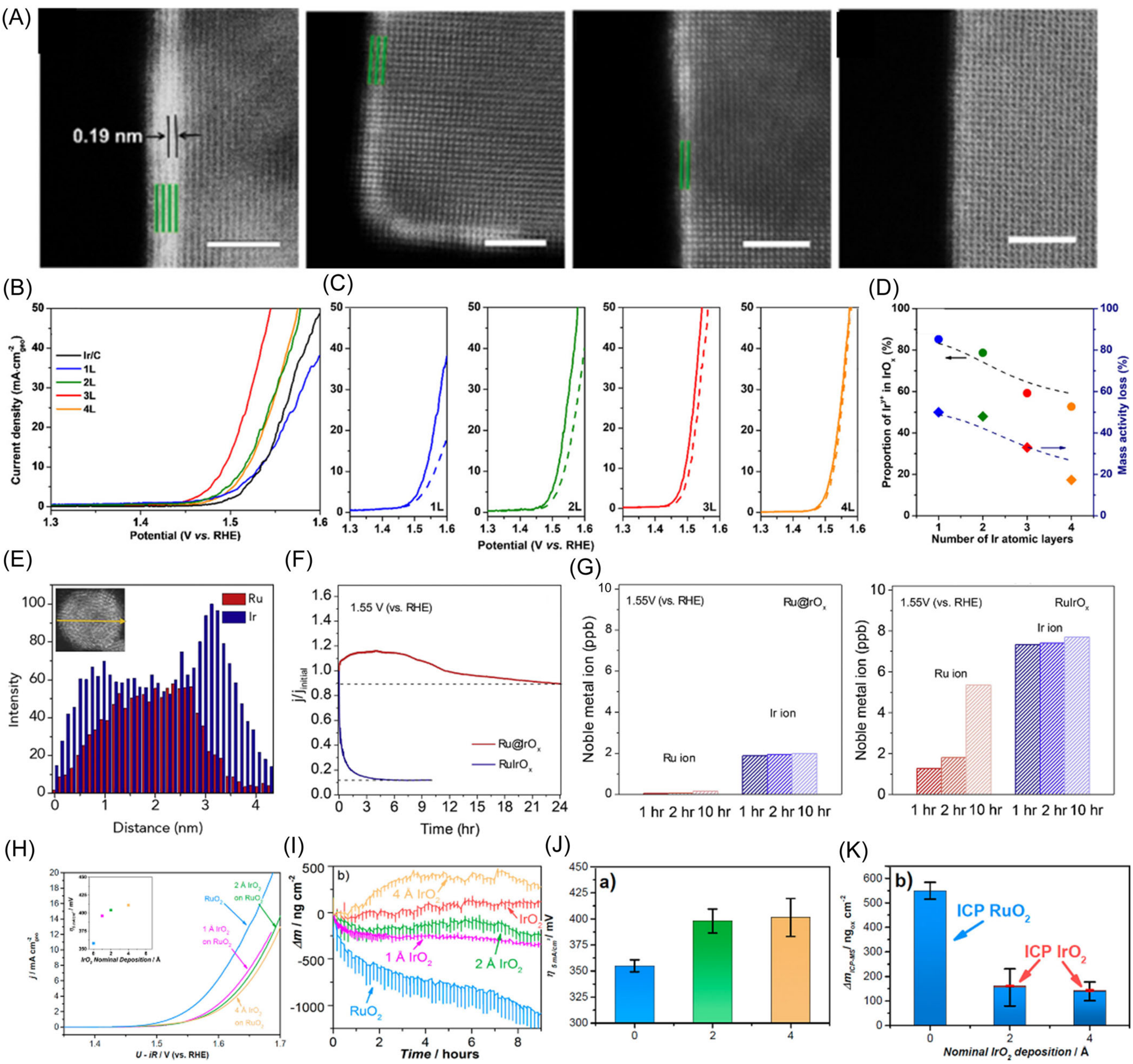

FIGURE 15 (A) Pd nanocubes coated with different thicknesses of Ir; (B) Polarization curves of various Ir-coated Pd nanocubes; (C) Variation in catalyst stability with increasing Ir thickness; (D) The proportion of unstable Ir in the surface iridium oxide, and the loss of catalyst mass activity; Reproduced with permission.[196] Copyright 2019, American Chemical Society; (E) EDX line profile of Ru@IrO$_x$ nanocrystals; (F) Chronocurrent curves of Ru@IrO$_x$ and RuIrO$_x$; (G) Concentrations of Ru and Ir ions in the electrolyte after different reaction times; Reproduced with permission.[197] Copyright 2019, Elsevier, Inc; (H) OER polarization curves of RuO_2 covered with different thicknesses of IrO$_x$; (I) Mass loss of catalyst over time; (J) Comparison of catalytic activity and (K) Mass loss covered with different thicknesses of IrO$_x$. Reproduced with permission.[198] Copyright 2018, American Chemical Society.

effectively reduced the adhesion energy of the graphene substrate to the Ir clusters. In particular, the coordination of Ir with pyridine N enabled it to be remarkably stabilized on the surface (Figure 16A). A B and N codoped graphene material has also been reported to serve as a substrate for IrO_2 nanoparticles for acidic water oxidation.[205] The doping of B and N in the carbon substrate could significantly enhance the SMSI, which ensured the effective anchoring of the IrO_2 nanoparticles, thereby avoiding the aggregation and shedding of the particles on the surface. Unfortunately, carbon-based materials as substrates are relatively unstable under acid OER conditions, which cannot meet the needs of long-term operation at high potential.

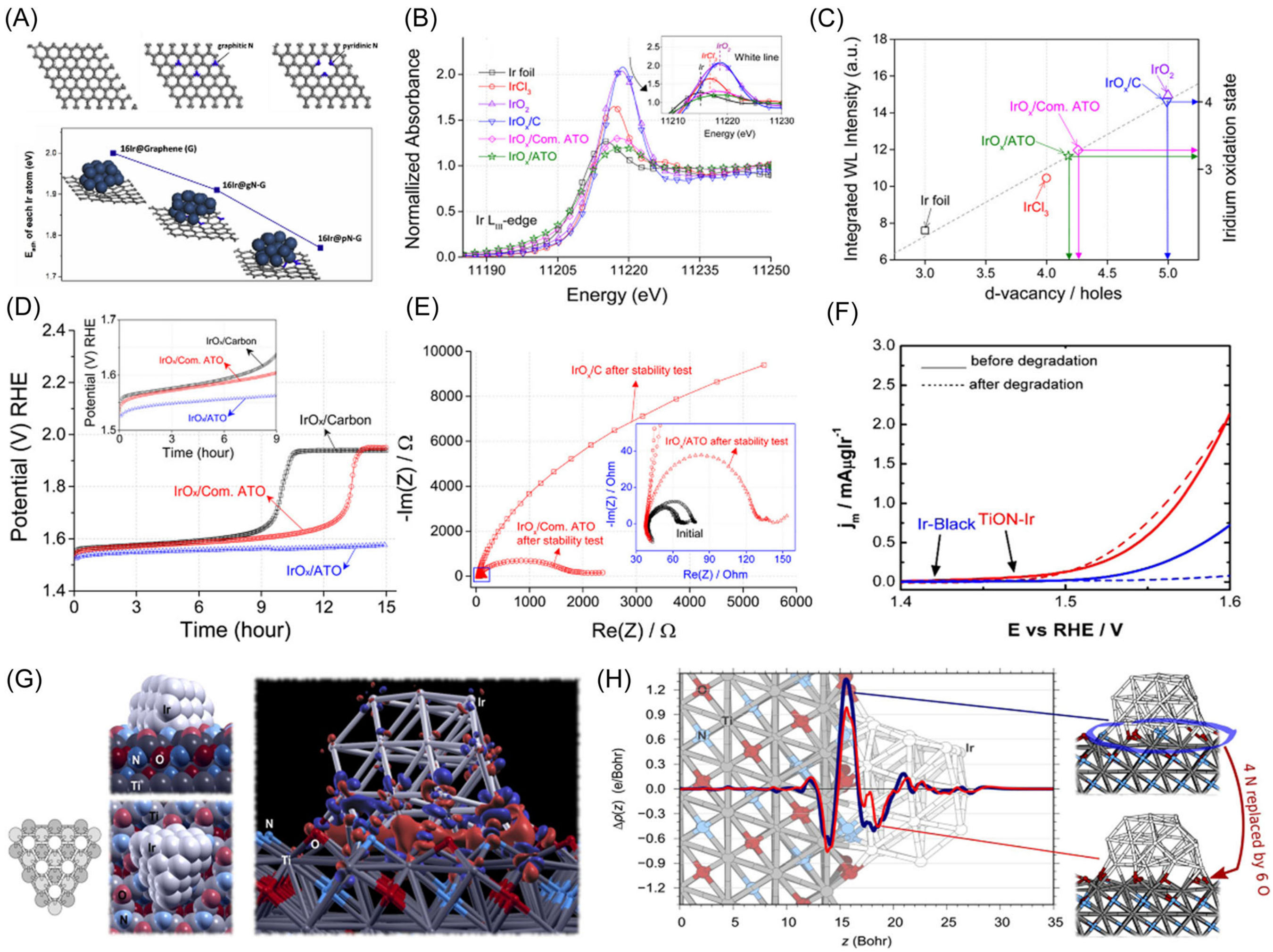

FIGURE 16 (A) Adhesion energies of pristine, pyridinic nitrogen and graphitic nitrogen-doped graphene to Ir; Reproduced with permission.[202] Copyright 2019, Elsevier, Inc; (B) Ir L III edge XANES of samples with different substrates; (C) Integrated white line (WL) with Ir oxidation state and D-vacancy; (D) Stability curves of IrO_x loaded on different substrates; (E) Impedance plots before and after the stability test; Reproduced with permission.[203] Copyright 2016, American Chemical Society; (F) LSV curves of TiON and carbon black loaded IrO_2 before and after accelerated cycling; (G) Electron charge density difference of Ir/TiON (red and blue represent electron accumulation and depletion, respectively); (H) Plane integral electron charge density difference. Reproduced with permission.[204] Copyright 2020, American Chemical Society.

As an alternative to carbon materials, metal oxides are regarded as promising substrate materials due to their good corrosion resistance and interaction with surface-supported noble metals.[206,207] For example, Nb-doped TiO_2 (ATO) could be tightly connected to the amorphous IrO_2 particles on its surface and had a small face resistance (Figure 16E).[203] Moreover, the ATO substrate enabled lower oxidizability of iridium on the surface (Figure 16B,C). The anchoring of the IrO_2 particles by the substrate significantly improved the acidic OER stability of the material (Figure 16D). The introduction of N into the oxide substrate was shown to better tune the SMSI effect. For example, Suhadolnik et al.[204] used nitrided TiO_2 films as substrates (TiON) to support Ir nanoparticles to explore the effect of N in the substrate on stabilizing Ir nanoparticles. The prepared TiON–Ir had stronger stability than carbon black supported Ir nanoparticles (Figure 16F). The electrons would transfer from the top Ti atoms and bottom Ir atoms to the O/N layer between them, which confirmed that nitrogen in the substrate produced adhesion to Ir nanoparticles (Figure 16G,H). Thus, this adhesion could also avoid the aggregation and shedding of surface particles during the OER process, which was beneficial to the maintenance of OER activity under operating conditions. Moreover, the follow-up work confirmed that the smaller particle size of the loaded nanoparticles would be beneficial to enhance the SMSI effect, resulting in higher stability.[208] In addition, it has been proposed that the interaction between catalyst and substrate can be enhanced by modification on the catalyst surface.[209] A Co_3O_4 catalyst coated with a carbon layer on a carbon cloth substrate was prepared by a two-step calcination method.[209] The sample without the surface carbon layer could continuously drive a current density of 100 mA cm^{-2} for 68.9 h, while the surface coating extended the operation time to 86.8 h. The authors claimed that this surface-coated carbon enhanced the coupling between the Co_3O_4 particles and the substrate material, thus avoiding deactivation due to active site shedding.

5.3 | Lattice oxygen route

Oxidation of lattice oxygen will lead to the formation of oxygen vacancies, which is also the crucial reason for excessive oxidation of metal atoms and deactivation of catalysts.[23] Therefore, avoiding the dissociation of lattice oxygen coordinated to the metal sites is the key to determining the stability of the catalyst.[23]

By constructing a stronger O coordination environment, precipitation of lattice oxygen can be suppressed, thereby maintaining stable performance.[210] The coordination of the introduced interstitial carbon and the lattice oxygen was proved to realize the prevention of the over-oxidation of the lattice oxygen. An interstitial carbon-introduced RuO_2 and RuSe composite (C–RuO_2–RuSe-10) was prepared by the evaporation of Se and the combustion of C using the $RuSe_2$ hollow nanospheres supported on carbon substrates as precursors (Figure 17A).[211] In contrast to RuO_2, the introduction of interstitial carbon in C–RuO_2–RuSe-10 led to the formation of C–O bonds (Figure 17B). Importantly, this bonding between the adjacent interstitial carbon and lattice oxygen further increased the difficulty of lattice oxygen dissociation. The theoretical calculations showed that the dissociation energy of lattice oxygen in the C–RuO_2–RuSe-10 was 0.8 eV higher than that in RuO_2, indicating the good stability of lattice oxygen in C–RuO_2–RuSe-10 (Figure 17D). Thus, C–RuO_2–RuSe-10 could operate continuously at a current density of 10 mA cm^{-2} for 50 h, exhibiting much stronger stability than RuO_2 (Figure 17C). Moreover, this C–O bond formation also increased the bond length of Ru–O, which led to a lower energy change of the RDS (*OOH formation) (Figure 17E). Therefore, the catalyst required an overpotential of 212 mV to drive a current density of 10 mA cm^{-2}, showing good catalytic performance.

Doping has also been confirmed to effectively increase the generation barrier of oxygen vacancies to improve the stability of the catalyst. A W and Er codoped RuO_2 ($W_{0.2}Er_{0.1}Ru_{0.7}O_{2-\delta}$) has been reported as an acidic OER catalyst with good activity and stability.[212] Compared with the O 2p band center (ε_p) (−3.31 eV) of RuO_2, the ε_p of $W_{0.2}Er_{0.1}Ru_{0.7}O_{2-\delta}$ was more negative (−4.12 eV) which suggested a larger difference between ε_p and Fermi level (Figure 17F,G). Moreover, the ΔG (2.29 eV) of $W_{0.2}Er_{0.1}Ru_{0.7}O_{2-\delta}$ to form V_O was also much higher than that of RuO_2 (0.67 eV) (Figure 17H). Therefore, the codoping of W and Er raised the generation barrier of oxygen vacancies, thereby suppressing the oxidation of lattice oxygen to oxygen. Due to the improved stabilization of lattice oxygen, the durability of $W_{0.2}Er_{0.1}Ru_{0.7}O_{2-\delta}$ was significantly enhanced, which could continuously drive a current density of 10 mA cm^{-2} for 500 h (Figure 17I). For nonprecious metal oxides, stabilization of lattice oxygen is also an effective strategy to promote catalyst stability. For example, the lattice oxygen in $CoMn_2O_4$, MnO_2, and Co_3O_4 was more easily oxidized to oxygen than that in Co_2MnO_4, which led to the destruction of the catalyst lattice.[79] The Mn atoms on the surface of Co_2MnO_4 could provide more electrons to the oxygen atom than the Co atoms in Co_3O_4 (Figure 17J,K). Therefore, the electron density on the surface of the O atom was higher, which confirmed the better stability of the Mn–O in Co_2MnO_4 (Figure 17L). As a result, the performance of this catalyst with a more stable lattice structure was maintained for a longer period of time, and Co_2MnO_4 could output a current density of 200 mA cm^{-2} for 1600 h (Figure 17M).

5.4 | Electronic structure regulation route

Precise control over the electronic structure is also an important aspect to achieve improved stability of acidic OER catalysts. Most materials are always easily oxidized and face severe deactivation problems under the electrochemical conditions of acidic OER. Therefore, the introduction of other electron-donating components to protect the active sites from excessive oxidation can avoid the rapid decay of catalytic activity.[137]

Electron transfer from the electron source to the active site is an important means to prevent excessive oxidation of the active center. For example, graphene has also been reported to act as an electron donor in acidic OER environments to slow the oxidation of metals.[213] The unique interface Ru centers between graphene and RuO_2 was obtained by moderate oxidation of the graphene-coated RuNi alloy (Figure 18A). This differential charge density at the interfacial Ru center showed that Ru and graphene at the interface showed accumulation and depletion of electrons, respectively. It indicated that the metal atoms in the outer layer of RuO_2 received electrons from the surface carbon layer, thereby slowing down the

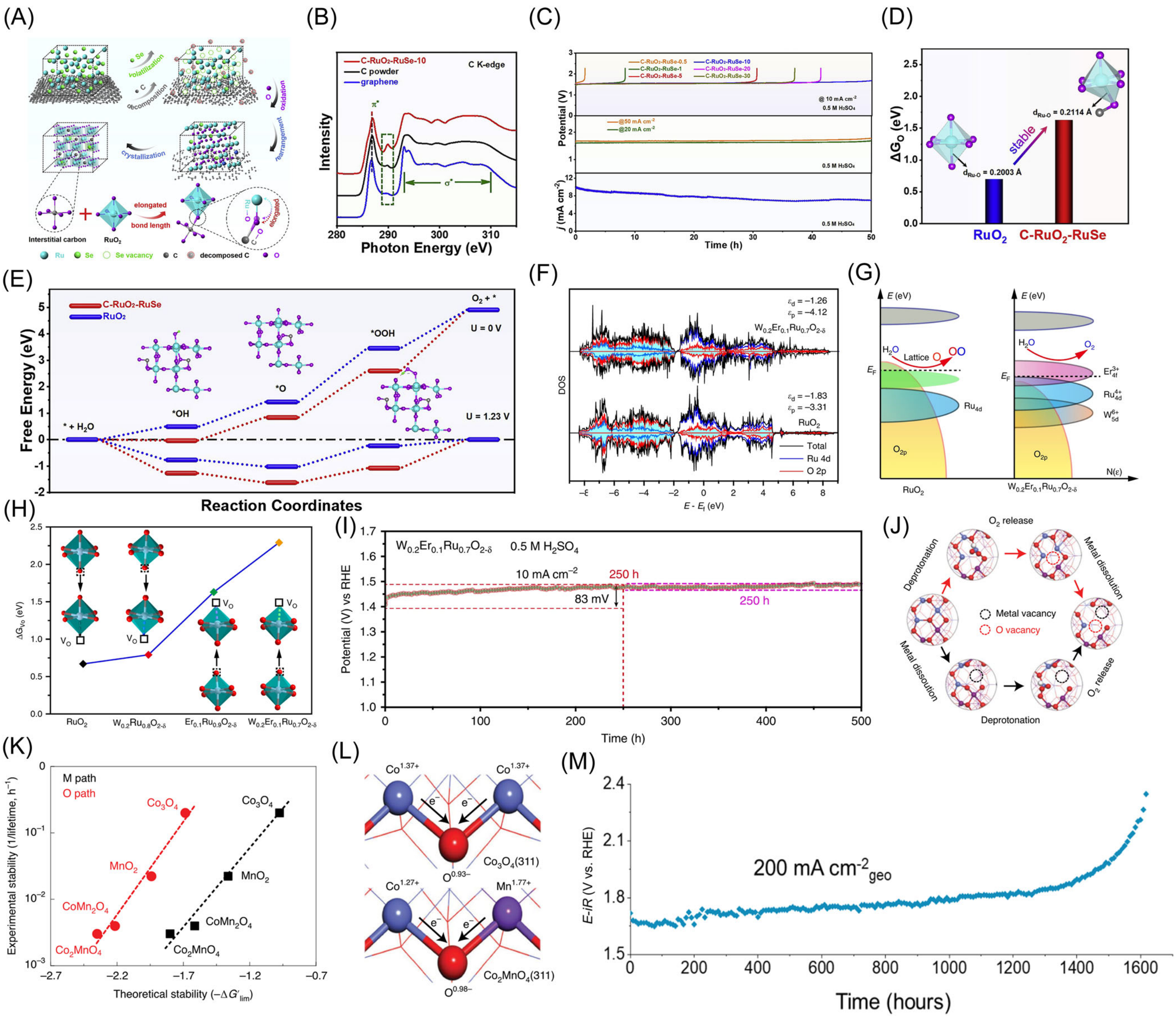

FIGURE 17 (A) Synthesis Scheme of C–RuO_2–RuSe-10; (B) XANES spectra of C K-edge C–RuO_2–RuSe-10; (C) Stability of C–RuO_2–RuSe obtained with different heating times; (D) ΔG_O of RuO_2 and C–RuO_2–RuSe-10; (E) Free energy changes of RuO_2 and C–RuO_2–RuSe-10; Reproduced with permission.[211] Copyright 2022, Elsevier, Inc; (F) DOS of $W_{0.2}Er_{0.1}Ru_{0.7}O_{2-\delta}$ and RuO_2; (G) Rigid band models of RuO_2 and $W_{0.2}Er_{0.1}Ru_{0.7}O_{2-\delta}$; (H) Oxygen vacancy formation energy of $W_{0.2}Er_{0.1}Ru_{0.7}O_{2-\delta}$ and RuO_2; (I) Stability test of $W_{0.2}Er_{0.1}Ru_{0.7}O_{2-\delta}$; Reproduced with permission.[212] Copyright 2020, Springer Nature; (J) Two lattice oxygen precipitation paths; (K) Stability of lattice oxygen in different oxides; (L) Lattice oxygen models in Co_2MnO_4 and Co_3O_4; (M) Stability test curve of Co_2MnO_4. Reproduced with permission.[79] Copyright 2022, Springer Nature.

process of excessive oxidation and degradation (Figure 18B). The tuning of the electronic structure allowed the catalyst to continuously drive a current density of 10 mA cm^{-2} for 24 h at the expense of a small reduction in activity (Figure 18C). Atom doping is also an efficient way to introduce electron donating components. For example, Pt-doped RuO_2 with a Pt core was prepared by oxidation of Ru-coated PtCo alloy as a precursor.[214] Due to the strong oxophilicity of Co during the oxidation process, the outward movement of Co made Pt more easily dispersed in RuO_2. With the progress of OER, Co atoms on the surface of PtCo–RuO_2/C would dissolve and form vacancies as active sites to drive the reaction. More importantly, the Pt on the surface of PtCo–RuO_2/C would be oxidized and provide electrons to Ru to avoid excessive oxidation and deactivation of Ru (Figure 18D,E). Therefore, the effective doping of Pt significantly protected the Ru sites, resulting in a significant improvement in stability. PtCo–RuO_2/C could continuously drive 10 mA cm^{-2} for 20 h without significant decay (Figure 18F).

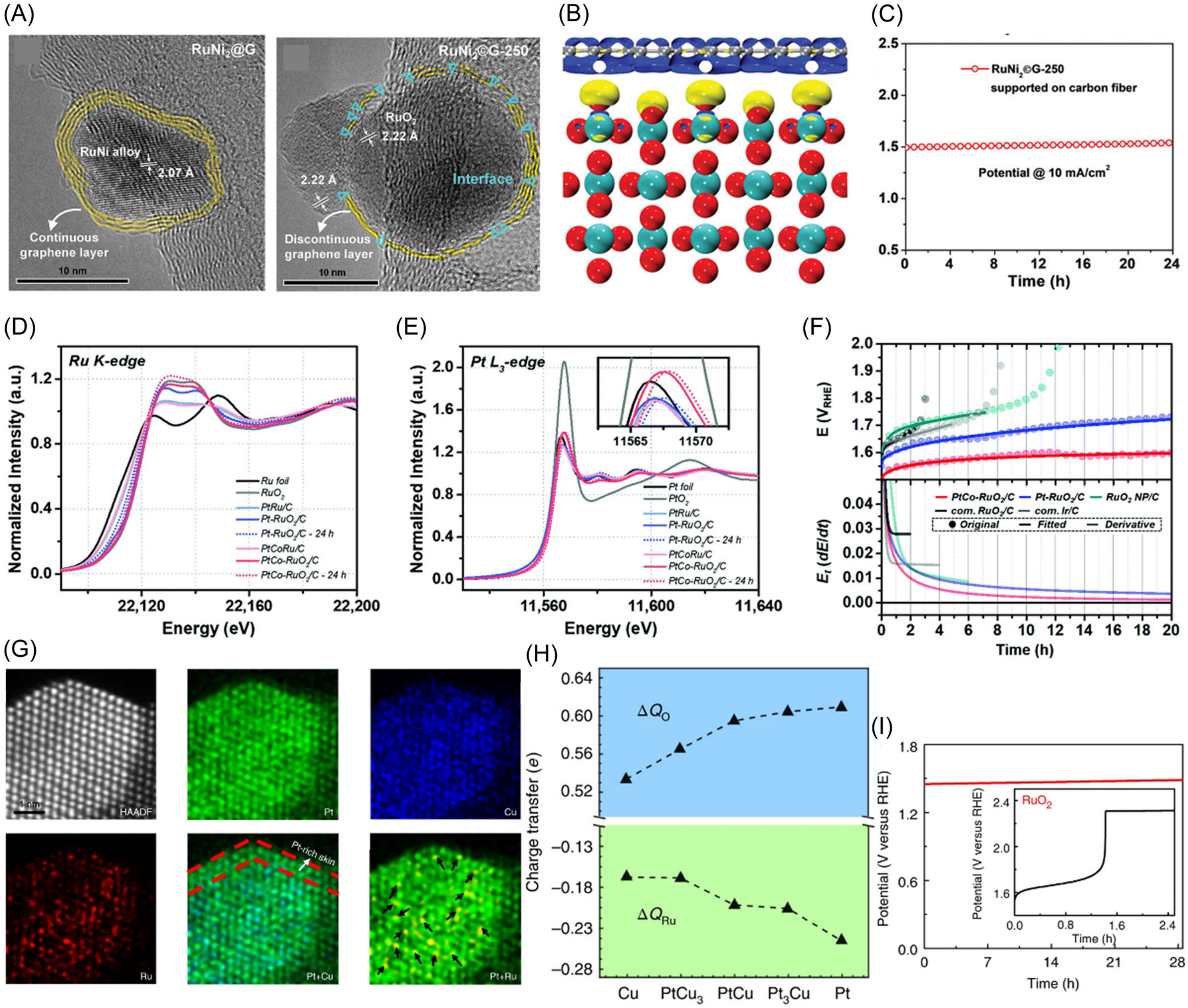

FIGURE 18 (A) HRTEM images of $RuNi_2$@G before and after oxidation; (B) Charge density at the interface between RuO_2 and graphene. (Blue and yellow represent charge depletion and accumulation, respectively); (C) Stability test of $RuNi_2$©G-250; Reproduced with permission.[213] Copyright 2020, John Wiley and Sons; (D) Ru K- and (E) Pt L_3-edge XANES region of catalysts before and after the OER test; (F) Stability curves of various catalysts; Reproduced with permission.[214] Copyright 2022, Royal Society of Chemistry; (G) HAADF image and corresponding element distribution; (H) Bader charge ΔQ for adsorbed oxygen and coordinated Ru; (I) Chronovoltometric diagram of Ru_1–Pt_3Cu. Reproduced with permission.[48] Copyright 2019, Springer Nature.

The introduction of the electron donating component has also been reported to act as an electron source to provide electrons to the reaction intermediates.[48] This can effectively avoid the occurrence of rapid oxidation due to losing too many electrons from the active site during the reaction. For example, Wu and coworkers loaded Ru atoms on $PtCu_x$/Pt skin core-shell structure as acidic OER catalysts (Figure 18G).[48] The PtCu alloy provided more charge to the adsorbed oxygen, while its coordination Ru_1 provided only less charge (Figure 18H). This suggested that PtCu substrates could act as a source of electrons for the reaction intermediates, thus slowing down the transfer of electrons from Ru to the oxygen-containing intermediates and thus avoiding their excessive oxidation. Besides, the defect-rich PtCu alloy substrate could effectively prevent the aggregation and shedding of Ru single atoms. As a result, this defect-rich PtCu alloy substrate with good electron-supplying ability anchored Ru single atoms with significantly enhanced acidic OER stability. The 97.9% of the original current density could be maintained after 28 h operation of Ru_1–Pt_3Cu at a current density of 10 mA cm^{-2} (Figure 18I).

5.5 | Self-healing route

Even the most stable catalysts are subject to dissolution under acidic OER conditions. These dissolved portions continue to act as active centers to drive the reaction through the self-healing process of redeposition.[89] Thus, the actual reaction process is actually a dynamic process of catalyst dissolution and healing.[215,216] The self-healing pathway can be used to create a dynamic and stable system that keeps the catalyst working.[217] The construction of appropriate lattices to achieve the combination of active sites and stable backbones is advantageous for the building of self-healing systems. For example, the Nocera group has constructed a $NiPbFeO_x$ that produced a unique self-healing phenomenon to achieve enhanced catalyst stability in acidic environments.[218] Nickel oxide acted as the active site to drive the reaction, and acid-stabilized Pb oxide acted as the backbone of the catalyst structure to support the redeposition of dissolved Ni. Due to the extreme susceptibility to corrosion, $NiFeO_x$ films were rapidly deactivated under acidic OER conditions. In contrast, $NiPbO_x$ and $NiPbFeO_x$ were able to drive currents of 1 mA cm^{-2} for up to 20 and 25 h at pH 2.5, respectively (Figure 19A). But the mechanism of the enhanced stability of Fe incorporation in acidic environments has not been explained. Additionally, Simonov et al.[219] showed that the self-healing phenomenon also existed in the $CoFePbO_x$ system under acidic OER conditions. Moreover, the appropriate addition of Co^{2+}, Fe^{3+}, and Pb^{2+} to the electrolyte was more favorable for the establishment of a dynamic equilibrium system, which further enhanced the stability of the catalyst (Figure 19B). However, this contamination or even destruction of the PEM will make the stable working of the electrolyzer challenging.

On the other hand, some of the catalysts loaded on the surface of the carrier can be reloaded onto the substrate after a redeposition process to continue driving the reaction. For example, a MnO_2 nanorod loaded with an array of Ru atoms was prepared to drive an acidic OER for a long period of time.[220] The prepared Ru/MnO_2 could output the current density of 100 mA cm^{-2} for 200 h, demonstrating relatively good stability (Figure 19C). As the reaction proceeded, the Mn content in the electrolyte rose rapidly and remained constant, while the Ru content maintained at a low level after a large fluctuation (Figure 19D). This demonstrated the dissolution of the unstable Mn and the redeposition of Ru in the Ru/MnO_2 catalyst under acidic OER conditions. After the self-healing process, the distribution of Ru arrays on MnO_2 changed from strips to sheets, thus favoring the catalytic activity improvement to some extent (Figure 19E,F). However, the redeposited Ru arrays tend to deposit on the table structure of MnO_2 and have a weaker integration with MnO_2, leading to a decrease in intrinsic activity. Therefore, although the activity of the deposited Ru array was slightly decreased, this self-healing process could effectively prevent Ru from dissolving and maintain the original activity of the catalyst (Figure 19G).

6 | CHALLENGES

The development of cost-effective acidic OER catalysts is the key to the widespread application of PEMWE. Despite the great achievements in acid OER in the past few years, there are still some challenges to be faced in the critical field:

1. OER, due to its inherent high potential, is always accompanied by many side reactions, such as oxidation of catalysts and support, current capacitance effects, and dissolution of catalysts, which may lead to overestimation of the catalytic activity in practical tests.
2. Although some current strategies have significantly improved the activity or stability of catalysts, the preparation of active and stable acidic OER catalysts is still difficult. Therefore, balancing the activity and stability of catalysts is necessary to meet the requirement of commercial applications.
3. The performance evaluation of catalysts using the standard three-electrode system in the laboratory is more limited, especially in practical applications where industrial-grade high currents and high temperatures are concerned.
4. In most of the previous studies on acidic OER catalysts, tests on activity and stability have only covered a small range of current densities. We call for future performance testing of catalysts at higher currents to meet the standards for industrial applications.
5. Due to the inherent high potential and strong corrosive environment of acidic OER, the composition and catalytic behavior of catalysts can be significantly altered. The utilization of in situ/operando characterization techniques is important to gain insight into the evolution of the catalyst structure and reaction mechanism.
6. During the OER process, many of the original catalysts undergo a reconfiguration process until they form a stable phase that can continue to drive the OER process. In contrast, most of the theoretical calculations involve only models of precatalysts and do not estimate the properties of the reconstituted material.

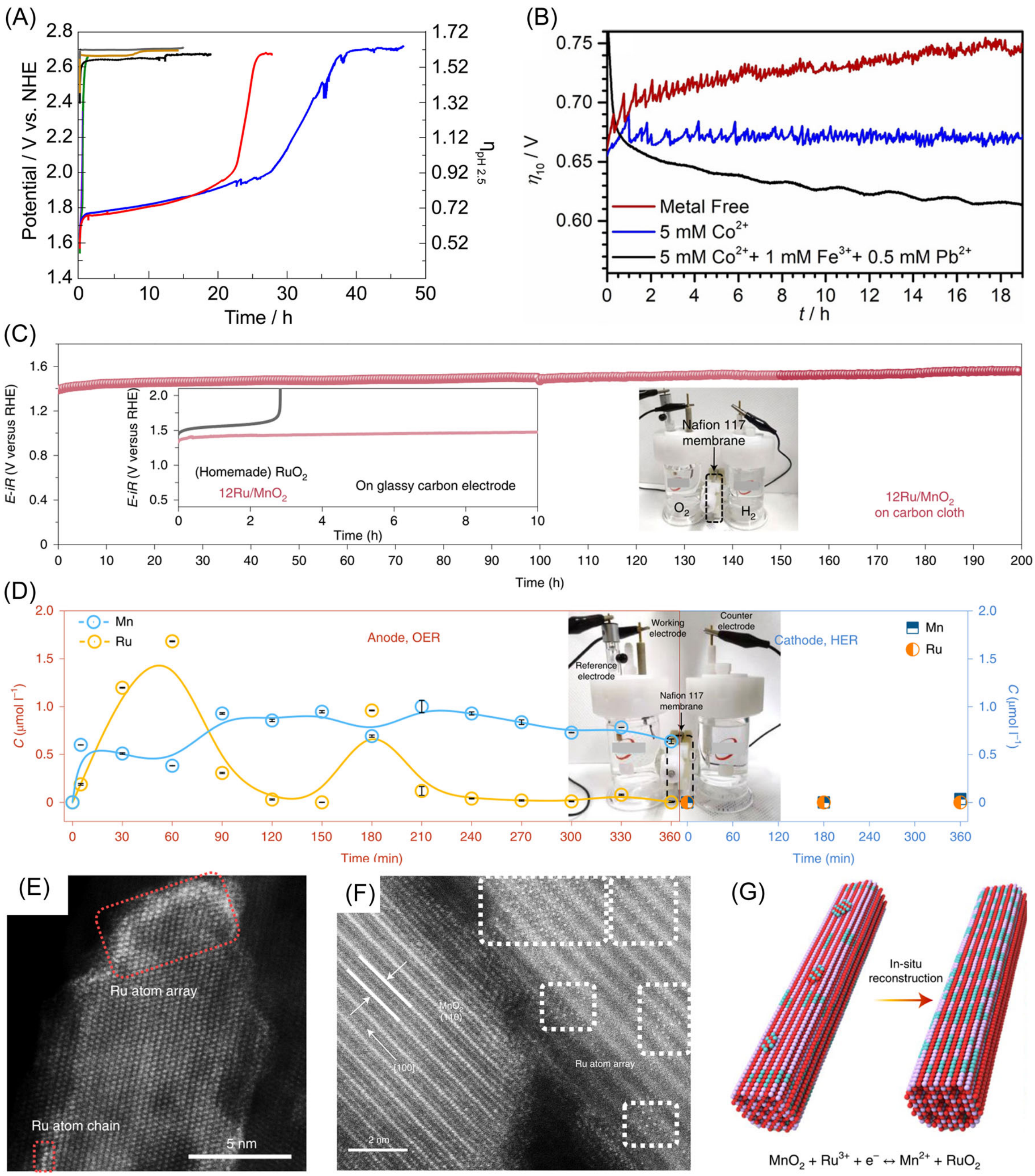

FIGURE 19 (A) Stability tests of $NiPbFeO_x$ (blue), $NiPbO_x$ (red), $PbFeO_x$ (mustard), $NiFeO_x$ (green) and PbO_x (gray). Reproduced with permission.[218] Copyright 2020, National Academy of Sciences; (B) Stability test of CoPbFeOx in electrolytes containing different amounts of metals. Reproduced with permission.[219] Copyright 2021, John Wiley and Sons; (C) Chronovoltometric tests of Ru/MnO_2; (D) Changes in the contents of Mn and Ru in the electrolyte during the Ru/MnO_2-driven OER process; (E, F) HADD-STEM images of Ru/MnO_2 before and after OER test; (G) Schematic diagram of Ru redeposition on Ru/MnO_2 surface. Reproduced with permission.[220] Copyright 2021, Springer Nature.

7. Although many studies have confirmed that LOM is involved in the OER process, most of the current theoretical calculations for material properties are still based on AEM only. This would lead to misleading and unreliable predictions or explorations about the true performance of the catalysts.

7 | CONCLUSIONS

The scale application of PEMWE depends on the development of acidic OER catalysts. The preparation of acidic OER catalysts with good stability and activity still faces many challenges. In this paper, the characteristics of the AEM and the LOM are summarized. The AEM tends to be a more stable but less active oxidation process, and the LOM shows excellent activity but is prone to catalyst degradation. Then we summarize three types of acidic OER catalysts, including precious metal-, nonprecious metal-, and carbon-based catalysts. To improve the activity of catalysts, some strategies are proposed in terms of composition, substrate, morphology, atom doping, defect, strain, and anion engineering. Finally, some routes to favor catalyst stability are suggested, including wrapping route, electronic coupling, stabilizing lattice oxygen, electronic structure regulation, and self-healing.

In the future, the development of new acid-stable and active OER catalysts is crucial for the development of PEMWE, which requires a combination of advanced characterization techniques, theoretical calculations, and specific experiments to gain a deep understanding of the dynamic changes and catalytic behavior during the reaction. In the end, we expect our review to provide some suggestions and guidance for the next design and development of acid-stabilized active OER catalysts.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

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